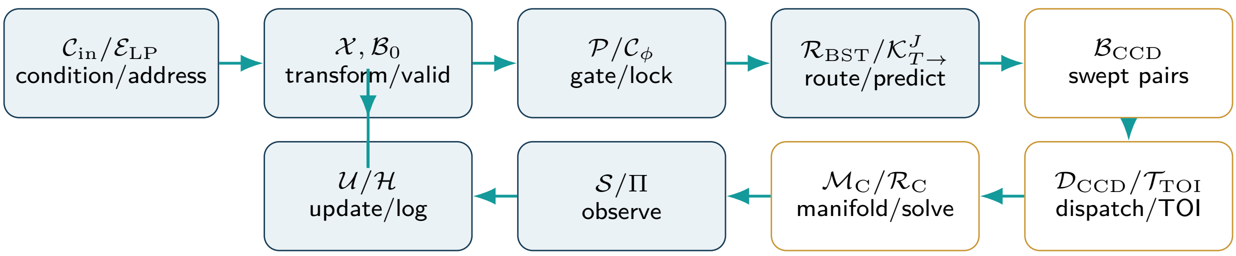
Non-Euclidean Holographic Data Fields

A Formal Operator Specification for a Parity-Conditioned Kinematic Foliation and Implicit Holographic Co-Processor



Core thesis

A valid memory state is represented as a point on a cell-local implicit zero-level set. A causal input conditioner and log-polar address select that state. After bounded routing and forward trajectory prediction, a general-purpose collision subsystem forms conservative swept pairs, dispatches by geometry and motion capability, localizes first contact as a certificate-carrying time interval, constructs simultaneous contact manifolds, and couples a declared response without crossing the boundary. Projection, feedback, and provenance close the self-referential engine.

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Formal specification, Version 0.3

31 August 2026

Document status: conceptual research specification and executable reference scaffold. It extends the supplied formalization with external primary research on literal computer-graphics CCD. It does not claim peer review, patent status, universal exactness, production validation, police endorsement, or application safety.

Author declaration and dedication

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Status of identity data	Included at the author's explicit request; not independently verified and not a cryptographic credential.

Dedication

Dedicated to police and emergency-service personnel who react, remain attentive to the public, and accept responsibility for public safety. The dedication expresses the author's intent. It does not imply adoption, validation, affiliation, or endorsement by any police organization.

The author-supplied identifier is bibliographic metadata only. A conforming implementation **MUST** not use a raw name, date of birth, or civil identifier as a cryptographic key, random seed, access token, or integrity secret. The provenance profile in this specification instead uses ordinary cryptographic key management, hash chaining, optional digital signatures, and privacy-preserving key identifiers.

Evidence boundary

The commute transcript contains time-stamped observations, personal impressions, location references, and generated technical claims. Version 0.3 retains the Version 0.2 evidence boundary and admits only ideas that can be stated as operators, invariants, hypotheses, or application profiles. The new CCD material is separately identified as external technical context. The transcript itself is not experimental validation, a chain-of-custody record, or proof that the proposed engine works.

Abstract

Version 0.1 established three linked artifacts. NHDF is a representation in which each valid lookup-table cell belongs to a non-degenerate local zero-level set. PCKF is the parity-conditioned dynamical law joining log-polar jitter addressing, forward-time kinematics, phase-accelerated branching, local boundary restoration, analytic sweeps, and feedback. IHCP is the bounded runtime that executes the declared operator order. Version 0.2 added the conformal log-polar metric, transform profiles, attention and interrupt control, a first conservative-advancement CCD profile, multimodal adapters, provenance, and responsible public-safety constraints.

Version 0.3 is a focused general-purpose Continuous Collision Detection extension. It replaces the single gap-function CCD block of Version 0.2 with a typed multi-backend contact subsystem: swept-volume broad phase; capability-based narrow-phase dispatch; analytic time-of-impact solvers for supported primitives; support-mapped convex shape casting; rigid-motion conservative advancement with angular and acceleration bounds; vertex-face and edge-edge CCD for linearly moving triangle meshes; interval/inclusion root isolation; certified implicit-field advancement; initial-overlap handling; contact-manifold construction; simultaneous-event grouping; self-collision filtering; deformable and topology-change profiles; event-driven integrator coupling; multi-contact response; and bounded parallel/GPU execution.

The term *general-purpose* is used deliberately but not universally. It means that one engine contract can cover rigid primitives, convex bodies, triangle and volume meshes, compounds, deformables, and valid implicit fields by dispatching to a backend whose assumptions are explicit. It does not mean that arbitrary discontinuous motion, an identically-zero field, non-Lipschitz geometry, unknown topology changes, or unsupported exact arithmetic can always yield a certified first contact. Every query returns a typed status and a time interval rather than a bare Boolean; unsupported or numerically indeterminate cases are exposed instead of silently accepted.

The resulting CCD claim is therefore literal and testable: under the declared motion, geometry, broad-phase enclosure, separation, arithmetic, tolerance, and termination contracts, the certified profile must not advance past the first contact. The log-polar LUT, parity gate, BST router, and other NHDF operators may prioritize, batch, or record contact work, but the no-tunneling guarantee comes from conservative geometric bounds and certified time-of-impact localization, not from those operators by themselves.

Revision 0.3 in one sentence

Version 0.3 turns the Version 0.2 contact guard into a backend-dispatched, certificate-carrying CCD subsystem that spans the common rigid, mesh, deformable, and implicit-geometry cases while exposing every unsupported or indeterminate case.

Revision summary

Area	What Version 0.2 lacked	Version 0.3 addition
Query contract	A point/reduced-configuration gap did not define object pairs, witnesses, features, or result certainty.	Typed pair query, time interval, minimal separation, witness points, feature identifiers, certificate class, and explicit status codes.
Motion	One relative-speed bound did not cover finite rigid bodies, rotation, acceleration, curved trajectories, or discontinuities.	Linear, rigid, polynomial, and deformable motion profiles; angular-radius and acceleration envelopes; mandatory interval splitting at discontinuities.
Geometry	No capability interface for primitives, convex shapes, meshes, compounds, volume elements, or approximate SDFs.	Shape proxy contract with swept bounds, support map, feature enumeration, distance/lower-bound oracle, Lipschitz/error data, and topology metadata.
Broad phase	No proof that all potentially colliding pairs reach the narrow phase.	Conservative swept AABB/BVH or declared continuous bounds, candidate completeness invariant, capacity and overflow policy, and self-collision adjacency filters.
Narrow phase	One generic SDF advance could not cover all common graphics geometry.	Analytic primitive solvers, convex shape casts, rigid conservative advancement, vertex-face and edge-edge mesh CCD, volume-element culling, implicit-field inclusion, and compound dispatch.
Numerics	Bracketing was named but exactness class, degeneracy handling, and floating-point policy were unspecified.	Interval/inclusion subdivision, filtered predicates, exact fallback, minimal separation, post-checks, deterministic tolerance scaling, and conservative terminal intervals.
Contacts	A single normal/velocity update did not handle simultaneous or persistent contacts.	Feature-stable contact records, co-time buckets, manifold clustering, contact islands, grazing state, persistence, and event queue rules.
Response	Point restitution omitted angular effective mass, multi-contact constraints, joints, resting contact, and initial overlap.	Impulse and friction equations with angular terms, bounded iterative manifold solver, barrier/contact-potential profile, overlap recovery, Zeno guard, and fail-safe freeze.
Deformation	No complete path for self-colliding or changing-topology meshes.	Linearized vertex trajectories, adjacency and normal-cone filters, surface/volume feature pairs, generation-tagged topology updates, and mandatory rebuild/invalidation rules.
Parallel runtime	No pair queues, backend batching, determinism policy, or GPU failure envelope.	Bounded stage queues, stable pair keys, backend sorting, per-query iteration caps, overflow spill/fail rules, deterministic replay profile, and certificate telemetry.
Verification	Thin-feature tests alone could not establish general-purpose correctness.	Adversarial primitive, convex, mesh, deformable, curved-motion, degeneracy, self-collision, broad-phase, CPU/GPU, and exact-ground-truth suites.

Contents

1

Paradigm identity, scope, and source boundary

1.1 Three names for three layers

The upgraded specification retains the layered identity introduced in Version 0.1.

Table 1.1: Layered identity of the paradigm.

Name	Role	Formal meaning
NHDF	Representation	A bounded lookup-table and active-cell model in which each valid state is embedded in a cell-local implicit zero-level set and may cross declared non-Euclidean chart transitions.
PCKF	Dynamics	A causal hybrid update law joining log-polar addressing, conformal metric compensation, parity and phase events, bounded branching, kinematics, boundary/contact control, and feedback.
IHCP	Runtime	A software or physical architecture that executes the typed operator chain, enforces resource and safety bounds, emits telemetry, and records the next state in a tamper-evident timeline.

1.2 Source materials

This document is grounded in three supplied artifacts.

- **[S0]** The original 30-page prompt-and-response transcript that introduced the chained vocabulary: parity, lower-case phi, log-polar jitter, kinematic calculus, an analytic cone, projections, distributed apex, BST/L-system branching, and a Klein-bottle/SDF-zero substrate.
- **[S1]** The 26-page *NHDF Formal Specification, Version 0.1*, which converted that vocabulary into a non-degenerate local-zero-set state machine and separated core claims from speculative extensions.
- **[S2]** The 40-page commute transcript, time-stamped approximately 07:50–09:03, which proposed missing scale encoding, extra transforms, continuous contact, attention, photometric and acoustic events, provenance, and public-safety applications.

The commute transcript is treated as a design notebook: useful ideas are admitted only after they receive a defined domain, codomain, equation or algorithm, resource bound, failure mode, and validation requirement.

1.3 Distinctive contribution

The paradigm is not any single ingredient. Signed distance fields, lookup tables, parity, L-systems, phase tracking, conservative advancement, hash chains, and log-polar coordinates are established techniques. The proposed contribution is their ordered closure under a common cell state and timeline:

1. input and causal history select a log-polar address;
2. the coordinate metric converts address-space motion into physical-space motion;
3. the selected candidate is mapped to a valid cell-local zero set;
4. parity, topology, phase, and resource conditions produce a bounded control event;

5. a branch router and forward integrator update distributed apices;
6. a contact guard prevents invalid time advancement under declared assumptions;
7. geometric or wavefront operators create an observable;
8. the observable and residual update the same state; and
9. the transition is appended to a tamper-evident causal record.

1.4 Normative core, profiles, and hypotheses

Table 1.2: Status classes used in Version 0.3.

Class	Included	Excluded until validated
Normative core	Typed state, local zero-set semantics, declared operator order, log-polar metric, parity meaning, causal time, bounded resources, feedback, failure exposure.	Any universal performance or robustness claim.
Conformance profile	Optional coordinate transforms, attention, collision, photometric, acoustic, provenance, public-safety, GPU, optical, or quantum adapter.	Equating profile inclusion with superiority.
Engineering hypothesis	Quantization advantage, recovery latency, contact efficiency, foveated savings, focus quality, tamper evidence, or operator-equivalent physical transfer.	Claims of proof before measurement against a baseline.
Prohibited inference	None.	Storm immunity, zero-cost computation, automatic error correction, automatic surveillance accuracy, autonomous coercive authority, or cryptographic proof from unsiged observations.

Evidence boundary

A profile name is not a performance result. An implementation may describe itself as conforming to a profile only when it satisfies the profile's semantics, bounds, telemetry, and failure conditions. It may claim an advantage only after a controlled comparison.

2

Semantic commitments and notation

2.1 Disambiguation of phi, angle, scale, and time

The sources overload lower-case phi. Version 0.3 retains the separated meanings.

Table 2.1: Core notation.

Symbol	Term	Definition
φ_g	Golden-ratio scalar	$\varphi_g = (1+\sqrt{5})/2$; an explicitly declared branch scale or priority parameter.
$\phi_i[n]$	Signal or cell phase	Wrapped or unwrapped phase associated with cell i at generation n.
θ	Polar direction	Geometric angle in a selected two-dimensional plane.
ρ_J	Log jitter magnitude	$\rho_J = \log(1+\gamma J)$; an address coordinate derived from residual magnitude.
w	Log spatial radius	$w = \log_b(r/r_0)$, with declared base $b > 1$, reference radius r_0, and finite clamp.
$\Delta^2\phi$	Phase acceleration	Second finite difference of phase, after declared unwrapping and filtering.
$T\rightarrow$	Forward timeline	A monotonically increasing logical generation with positive elapsed time and no future-state read.
$F_i(z,t) = 0$	Local validity condition	Cell i's candidate state lies on a nontrivial local zero-level set.
K	Klein-bottle topology	An intrinsic quotient/chart-transition rule, not a claim that one global Euclidean SDF exists.

2.2 Required non-degenerate reading of SDF zero

If one global scalar field is identically zero, then its gradient vanishes wherever it is differentiable. It supplies no distance, normal, interior/exterior label, or restoring direction. Version 0.3 therefore retains the distributed reading:

$$F_i : Z_i \times \mathbb{T} \to \mathbb{R}, \tag{2.1}$$

$$M_i(t) := \{z \in Z_i : F_i(z,t) = 0\}, \tag{2.2}$$

$$z_i(t) \in M_i(t) \qquad \text{for each active cell } i. \tag{2.3}$$

A field called an SDF **MUST** satisfy its declared distance approximation and gradient conditions. Otherwise it is an implicit constraint or residual field and is named as such.

2.3 Discrete time as reference semantics

The normative timeline is

$$n \in \mathbb{N}, \qquad T_{n+1} > T_n, \qquad \Delta t_n = T_{n+1} - T_n > 0. \tag{2.4}$$

Every state-bearing buffer slot carries a logical generation. Physical ring-buffer wrap does not reverse time. A transition computing generation $n+1$ may read only generations at most n.

2.4 Cell state

A Version 0.3 cell is the typed record

$$c_i = \big(q_i,\, u_i,\, a_i,\, v_i,\, \alpha_i,\, \phi_i,\, \Delta\phi_i,\, \Delta^2\phi_i,\, \kappa_i,\, p_i,\, o_i,\, \nu_i,\, b_i,\, \chi_i\big), \tag{2.5}$$

where q_i is the packed payload; u_i the log-polar/time address; a_i, v_i, α_i apex position, velocity, and acceleration; κ_i the ordering key; p_i a parity event; $o_i \in \{+1, -1\}$ a local orientation; ν_i links, chart, epoch, and allocation metadata; b_i a resource/attention weight; and χ_i a profile and quality-flag word.

2.5 The apex is a field, not a singular global point

The active apex set is

$$\mathcal{A}_n = \{a_i[n] : i \in I_n\}. \tag{2.6}$$

A weighted centroid may be observed, but no single apex is the sole anchor. The log-polar origin requires a finite implementation rule, developed in chapter 5.

3

Operator algebra and canonical closure

3.1 Typed operators

Table 3.1: Normative and profile operator signatures.

Operator	Signature	Role
$\mathcal{C}_{\text{in}}$	$X \times H \to X'$	Sensor calibration, photometric normalization, observer-frame compensation, and quality flags; profile-dependent.
$\mathcal{E}_{\text{LP}}$	$X' \times H \to \Omega_{\rho\theta T}$	Causal residual, log magnitude, polar direction, forward-time address, and saturation status.
$\mathcal{A}_B$	$\Omega \times L \to \mathcal{I}_B$	Budgeted attention/resource selection with a nonzero peripheral monitoring floor.
$\mathcal{X}_\xi$	$Z \to Z$	Declared optional coordinate transform selected from a typed profile: identity, Möbius inversion, log-spiral shear, or hyperbolic transport.
$\mathcal{B}_0$	$Z_i \to M_i$	Project or embed a candidate on a cell-local zero set within tolerance.
$\mathcal{P}$	$\mathbb{B}^w \times \{\pm 1\} \to \mathbb{B}^k$	Data parity, topology orientation parity, or diagnostic pair; $k = 1$ preserves the source constraint.
$\mathcal{C}_\phi$	$\Phi \times \mathbb{B}^k \to C$	Phase lock, debounce, hysteresis, orientation-chart transition, and safe-mode request.
$\mathcal{R}_{\text{BST}}$	$C \times B \times \mathbb{R} \to \mathcal{T}$	BST ordering and L-system bifurcation at declared branch nodes under capacity bounds.
$\mathcal{K}^J_{T\to}$	$\mathcal{T} \times \Delta t \to \widehat{\mathcal{T}}$	Predict forward trajectories using the declared log-polar Jacobian, physical acceleration, damping, and generation semantics.
$\mathcal{B}_{\text{CCD}}$	$\widehat{\mathcal{T}} \times \mathcal{G} \to \mathcal{Q}$	Conservatively produce candidate object/feature pairs whose swept bounds may overlap.
$\mathcal{D}_{\text{CCD}}$	$\mathcal{Q} \times \mathcal{K} \to \mathcal{Q}_D$	Select a narrow-phase backend from motion and geometry capabilities; reject unsupported combinations explicitly.
$\mathcal{T}_{\text{TOI}}$	$\mathcal{Q}_D \to \mathcal{E}_C$	Compute a certified or conservative time-of-impact interval, or an exposed indeterminate/failure result.
$\mathcal{M}_{\text{C}}$	$\mathcal{E}_C \to \mathcal{M}_C$	Group co-time contacts, construct feature-stable witness records, and build contact islands/manifolds.
$\mathcal{R}_{\text{C}}$	$\widehat{\mathcal{T}} \times \mathcal{M}_C \to \mathcal{T}'$	Couple detection to a declared impulse, constraint, barrier, or application response without changing the detection certificate.
$\mathcal{C}_{\text{GP}}$	$\widehat{\mathcal{T}} \times \mathcal{G} \to \mathcal{T}'$	Composite general-purpose contact operator $\mathcal{R}_{\text{C}} \circ \mathcal{M}_{\text{C}} \circ \mathcal{T}_{\text{TOI}} \circ \mathcal{D}_{\text{CCD}} \circ \mathcal{B}_{\text{CCD}}$.
$\mathcal{S}$	$\mathcal{T}' \to G$	Analytic cone, implicit geometry, acoustic delay, or other declared wavefront/field construction.
Π	$G \to Y$	Image, projection, beam, field slice, feature, or control observable.

Operator	Signature	Role
$\mathcal{U}$	$L \times Y \times R \to L$	Feedback, allocation, residual, quality, and generation update.
$\mathcal{H}$	$E \times h \to h'$	Canonical event digest and append-only provenance-chain update.

3.2 Expanded canonical chain

Let L_n be the complete bounded state and h_n the provenance-chain head. Kinematics first predict a trajectory family $\widehat{\mathcal{T}}_n$. The general-purpose CCD block then restricts that prediction to a non-crossing state or reports failure:

$$\begin{aligned}
\widehat{\mathcal{T}}_n &= \mathcal{K}^J_{T\to} \circ \mathcal{R}^{\varphi_g,\Delta^2\phi_n}_{\mathrm{BST}} \circ \mathcal{C}_\phi \circ \mathcal{P} \circ \mathcal{B}_0 \circ \mathcal{X}_{\xi_n} \circ \mathcal{A}_B \circ \mathcal{E}_{\mathrm{LP}} \circ \mathcal{C}_{\mathrm{in}}(x_n; L_n), && (3.1)\\
\mathcal{T}'_n &= \mathcal{C}_{\mathrm{GP}}(\widehat{\mathcal{T}}_n; \mathcal{G}_n), && (3.2)\\
y_n &= \Pi \circ \mathcal{S}(\mathcal{T}'_n), && (3.3)\\
L_{n+1} &= \mathcal{U}(L_n, y_n, r_n), && (3.4)\\
h_{n+1} &= \mathcal{H}(h_n, E_{n+1}). && (3.5)
\end{aligned}$$

Equivalently, the observable path is

$$y_n = \Pi \circ \mathcal{S} \circ \mathcal{R}_{\mathrm{C}} \circ \mathcal{M}_{\mathrm{C}} \circ \mathcal{T}_{\mathrm{TOI}} \circ \mathcal{D}_{\mathrm{CCD}} \circ \mathcal{B}_{\mathrm{CCD}} \circ \mathcal{K}^J_{T\to} \circ \mathcal{R}_{\mathrm{BST}} \circ \mathcal{C}_\phi \circ \mathcal{P} \circ \mathcal{B}_0 \circ \mathcal{X}_{\xi_n} \circ \mathcal{A}_B \circ \mathcal{E}_{\mathrm{LP}} \circ \mathcal{C}_{\mathrm{in}}(x_n; L_n). \quad (3.6)$$

The coordinate transform $\mathcal{X}_{\xi_n}$ is the identity unless a conformance profile declares another transform, its trigger, and its inverse or fallback. The general-purpose CCD block is not optional for an implementation claiming the Version 0.3 CCD profile.

3.3 Normative precedence

Unless an implementation proves semantic equivalence of a fused stage, the order is:

1. acquire, calibrate, tag quality, and stabilize the observer frame;
2. derive causal residuals and log-polar/time addresses;
3. select active cells under a finite attention and work budget;
4. apply the declared coordinate-transform block;
5. project candidates to local zero sets;
6. compute data/topology parity and phase-control state;
7. perform bounded branch routing;
8. predict forward object, feature, and boundary trajectories;
9. form conservative swept bounds and a complete candidate-pair set;
10. dispatch each candidate to a backend whose capability contract covers the pair;
11. localize the earliest contact as a certified or conservative time interval;
12. combine contacts within the declared co-time tolerance into manifolds and islands;
13. apply the declared response or barrier coupling, or expose an indeterminate/failure state;
14. construct wavefront/implicit geometry and project an observable;
15. feed the residual and output into the next bounded state; and
16. append a canonical event digest to the causal provenance chain.

Normative requirement

Changing operator order is a semantic change, not an optimization detail. In particular, collision response may not retroactively justify a missed time of impact, and a broad phase may not discard a pair unless its swept bound is conservative. A reordered or fused implementation **MUST** publish the new composition and show, with reference vectors, when it is equivalent and when it is not.

3.4 Result monotonicity and failure exposure

The CCD block returns more than a Boolean. A query result contains

$$R_C = (s,\ [\underline{t}, \overline{t}],\ d_{\min},\ i, j,\ f_i, f_j,\ x_i, x_j,\ n,\ \gamma,\ k,\ \chi), \tag{3.7}$$

where s is a status, $[\underline{t}, \overline{t}]$ is a time-of-impact enclosure, $d_{\min}$ is the requested minimal separation, (i, j) are object identifiers, (f_i, f_j) feature identifiers, (x_i, x_j) witness points, n a contact normal when defined, γ the certificate class, k an iteration/work count, and χ diagnostic flags.

For a fixed query and increasing work budget, a certified enclosure should be monotone: the interval may shrink but may not exclude the true first contact. A backend that cannot maintain this property returns `INDETERMINATE` or a weaker, clearly labelled approximate result; it does not silently return `CLEAR`.

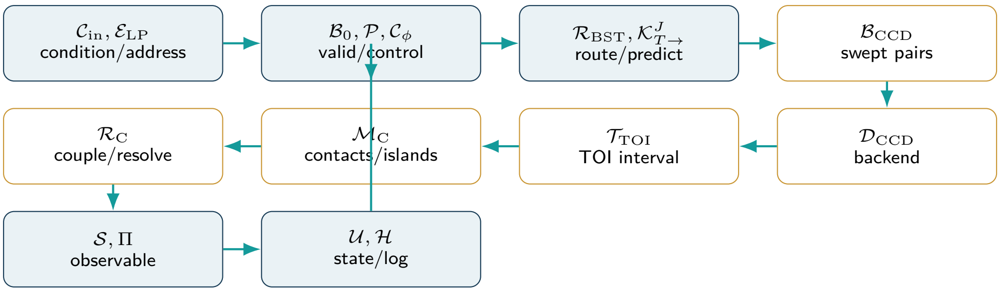

Figure 3.1: Version 0.3 operator chain. Gold stages form the general-purpose CCD subsystem.

4

Local zero-set substrate and topology

4.1 Boundary projection

For a differentiable cell-local field with nonzero gradient near its zero set, one regularized projection step is

$$\mathcal{B}_0(z) = z - \lambda_B \frac{F_i(z, T_n)}{\|\nabla F_i(z, T_n)\|_2^2 + \varepsilon_B} \nabla F_i(z, T_n), \tag{4.1}$$

with $0 < \lambda_B < 2$ and $\varepsilon_B > 0$. The residual is

$$r_{B,i} = |F_i(z_i, T_n)|. \tag{4.2}$$

An implementation uses a finite tolerance $r_{B,i} \leq \tau_B$ and records singular or ill-conditioned gradients.

4.2 Klein-bottle chart rule

The intrinsic Klein bottle is represented by the quotient

$$(0, v) \sim (1, 1 - v), \qquad (u, 0) \sim (u, 1). \tag{4.3}$$

Crossing the first seam reverses a chart coordinate and therefore flips a local orientation sign. This supplies a concrete topological event; it does not supply error correction or a globally signed Euclidean distance.

4.3 Data parity and topology parity

For payload $q_i \in \mathbb{B}^w$,

$$p_i^{\text{data}} = \bigoplus_{b=0}^{w-1} \text{bit}_b(q_i). \tag{4.4}$$

For a chart path Γ_i with transition orientations $\sigma_e \in \{+1, -1\}$,

$$p_i^{\text{topo}} = \frac{1 - \prod_{e \in \Gamma_i} \sigma_e}{2}. \tag{4.5}$$

The one-bit source-compatible gate may be $p_i = p_i^{\text{data}} \oplus p_i^{\text{topo}}$. A diagnostic profile stores both bits separately so that corruption and orientation events are not conflated.

Evidence boundary

XOR parity detects an odd number of inversions between checks, misses every even number, cannot locate the damaged bit, and cannot repair it. Any integrity claim requires stronger coding, redundancy, recomputation, or an application residual.

5

Conformal log-polar kinematics

5.1 Why the scale metric is required

The commute transcript correctly identifies a missing element in a bit-shift interpretation of log-polar motion: one address step does not correspond to one constant physical displacement. Let

$$w = \log_b(r/r_0), \qquad r = r_0 b^w, \qquad b > 1, \tag{5.1}$$

and let θ be the geometric polar angle. In a two-dimensional plane,

$$p(w,\theta) = p_0 + r_0 b^w \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}. \tag{5.2}$$

The Jacobian is

$$J_{\mathrm{LP}}(w,\theta) = r \begin{bmatrix} (\ln b)\cos\theta & -\sin\theta \\ (\ln b)\sin\theta & \cos\theta \end{bmatrix}. \tag{5.3}$$

For natural logarithms, $\ln b = 1$ and $J_{\mathrm{LP}} = rR(\theta)$. For base two, the radial derivative carries the factor $\ln 2$ even if an integer implementation uses shifts to synthesize 2^w.

The induced metric is

$$G_{\mathrm{LP}} = J_{\mathrm{LP}}^{\mathsf{T}} J_{\mathrm{LP}} = r^2 \operatorname{diag}\left((\ln b)^2, 1\right), \tag{5.4}$$

and the area scale is

$$|\det J_{\mathrm{LP}}| = (\ln b) r^2. \tag{5.5}$$

Consequently, an equal increment in w or θ has a radius-dependent physical meaning. This scale factor is the rigorous version of the transcript's "radial magnitude" or "Jacobian" correction.

5.2 Velocity and acceleration

Let $e_r = (\cos\theta, \sin\theta)^{\mathsf{T}}$ and $e_\theta = (-\sin\theta, \cos\theta)^{\mathsf{T}}$. The physical velocity is

$$\dot{p} = r\left((\ln b)\dot{w}\, e_r + \dot{\theta}\, e_\theta\right). \tag{5.6}$$

The physical acceleration is

$$\ddot{p} = r\left(\left((\ln b)\ddot{w} + (\ln b)^2\dot{w}^2 - \dot{\theta}^2\right)e_r + \left(\ddot{\theta} + 2(\ln b)\dot{w}\dot{\theta}\right)e_\theta\right). \tag{5.7}$$

The $-\dot{\theta}^2$ term is centripetal and the $2(\ln b)\dot{w}\dot{\theta}$ term is the radial-angular cross coupling that an independent barrel shift omits. A conforming kinematic profile **MUST** either evaluate these terms, use a numerically equivalent Cartesian update, or declare a bounded approximation and its error.

5.3 Discrete update

For a cell with log-polar state $\zeta_i = (w_i, \theta_i)$ and physical state (a_i, v_i), two valid implementation strategies are:

1. integrate a_i, v_i in Cartesian space and encode the result back into (w_i, θ_i); or
2. integrate ζ_i using the metric, Christoffel terms, and a declared manifold integrator.

The reference implementation uses Cartesian semi-implicit integration:

$$v_i[n+1] = v_i[n] + \Delta t_n\, \alpha_i[n], \tag{5.8}$$

$$a_i[n+1] = a_i[n] + \Delta t_n\, v_i[n+1], \tag{5.9}$$

$$(w_i[n+1], \theta_i[n+1]) = \mathrm{encode}_{\mathrm{LP}}(a_i[n+1] - p_0). \tag{5.10}$$

This avoids pretending that an address shift alone is a complete physical integrator.

5.4 Quantization and radial error

Let $\widehat{w} = Q_w(w)$ with maximum log-coordinate error $|\widehat{w} - w| \leq \Delta w/2$. The relative radial error obeys

$$\left| \frac{\widehat{r} - r}{r} \right| \leq b^{\Delta w/2} - 1. \tag{5.11}$$

Angular quantization with $|\widehat{\theta} - \theta| \leq \Delta\theta/2$ contributes approximately $r\Delta\theta/2$ tangential error at small angles. Version 0.3 therefore requires calibration of both radial and angular quantizers rather than assuming that a fixed bit width is uniformly accurate.

5.5 Apex regularization

The polar apex $r = 0$ corresponds to $w = -\infty$ and cannot be represented in finite registers. A conforming implementation **MUST** declare one of the following:

- a finite clamp $r \geq r_{\min} > 0$, with $w_{\min} = \log_b(r_{\min}/r_0)$;
- an apex sentinel with a separate local Cartesian micro-chart; or
- a multi-chart atlas that excludes the singular coordinate.

Saturation at $w_{\min}$ is logged. No claim of infinite foveal resolution is permitted on finite hardware.

Correction to a commute claim

Moving inward or outward on a log-polar grid changes spatial resolution and quantization. It does not, by itself, define physical acceleration or deceleration. Acceleration remains the derivative of physical velocity and must obey the Jacobian-transformed law above.

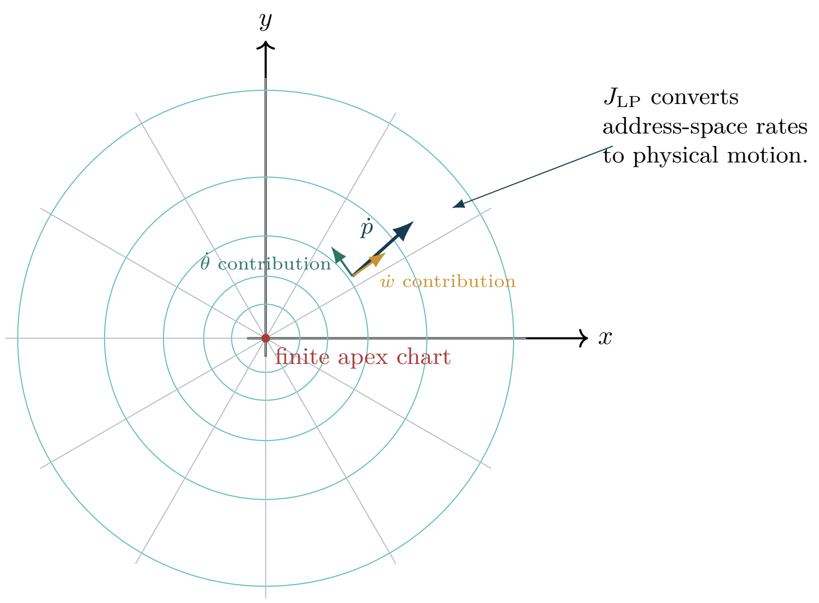

Figure 5.1: Log-polar cells expand with radius. The Jacobian and cross terms are part of the dynamics, not optional visual decoration.

6

Declared coordinate-transform profiles

6.1 Transform block semantics

The commute transcript introduces several named operators as if each were a bitwise instruction. Version 0.3 admits them only as typed coordinate transforms. The transform block is

$$\mathcal{X}_\xi = \mathcal{G}_c \circ \Gamma_\kappa \circ \mathcal{M}_{c,R} \tag{6.1}$$

when all three profiles are enabled; disabled stages are the identity. This order is canonical: inversion first establishes scale/orientation, log-spiral shear then couples radial and angular coordinates, and geodesic transport finally translates in the declared metric. An implementation may choose a different order only by publishing it as a distinct profile.

6.2 Möbius inversion profile

Represent a planar coordinate by $z = x + iy$. The source form $z \mapsto 1/z$ is generalized to

$$\mathcal{M}_{c,R}(z) = c + \frac{R^2}{z - c}, \qquad z \neq c. \tag{6.2}$$

For $c = 0$, $R = 1$, and $z = re^{i\theta}$,

$$r' = r^{-1}, \qquad \theta' = -\theta, \tag{6.3}$$

so in log-polar coordinates

$$w' = -w, \qquad \theta' = -\theta \pmod{2\pi}. \tag{6.4}$$

With a general inversion radius, $w' = 2\log_b(R/r_0) - w$. The map is involutive away from its pole.

A local field transforms by pullback:

$$F_i^{\mathcal{M}}(z,t) = F_i(\mathcal{M}_{c,R}^{-1}(z), t). \tag{6.5}$$

Thus the zero set is transformed without claiming that the field values remain exact signed distances.

Register-equivalence requirement

A bitwise NOT is not generally Möbius inversion. For a fixed-point register $q = \mathrm{enc}(w)$, an implementation **MUST** show that its register operation satisfies $\mathrm{dec}(M_q(q)) = -\mathrm{dec}(q) + c$ within tolerance. In two's-complement arithmetic, negation is normally $\sim q + 1$, not merely $\sim q$.

6.3 Log-spiral shear profile

The commute transcript's "geodesic ray-twister" is made concrete as an invertible shear in log-polar coordinates:

$$\Gamma_{\kappa,\delta}(w,\theta) = (w + \delta_w,\ \mathrm{wrap}(\theta + \kappa w + \delta_\theta)). \tag{6.6}$$

Its Jacobian in (w,θ) is

$$D\Gamma = \begin{bmatrix} 1 & 0 \\ \kappa & 1 \end{bmatrix}, \qquad \det D\Gamma = 1. \tag{6.7}$$

Lines of constant transformed phase correspond to logarithmic spirals in physical space. A dynamic twist may be driven by a bounded second phase difference,

$$\kappa_i[n] = \mathrm{clip}(\kappa_0 + k_\kappa \Delta^2 \phi_i[n], -\kappa_{\max}, \kappa_{\max}). \tag{6.8}$$

The shear couples scale and direction but does not replace physical frame compensation or contact detection.

6.4 Optional hyperbolic gyrotranslation

If, and only if, the state is explicitly mapped into a Poincaré ball $\mathbb{D}_c^d = \{x : c\|x\|^2 < 1\}$ of curvature $-c$, $c > 0$, a gyrovector translation may use Möbius addition

$$u \oplus_c v = \frac{(1 + 2c\langle u, v\rangle + c\|v\|^2)u + (1 - c\|u\|^2)v}{1 + 2c\langle u, v\rangle + c^2\|u\|^2\|v\|^2}. \tag{6.9}$$

The operator $\mathcal{G}_c(u; v) = u \oplus_c v$ is a research profile. It is invalid to apply the formula to arbitrary packed words without an encode/decode map, norm bound, pole handling, and reference implementation.

6.5 Composition, inverse, and quality flags

Every transform profile **MUST** publish:

- domain, codomain, parameters, and singular sets;
- forward map and, where required, inverse map;
- effect on local fields, gradients, and orientation;
- quantization and overflow behavior;
- a quality flag indicating saturation, pole proximity, or inverse failure; and
- reference vectors for identity, boundary, and extreme cases.

7

Input conditioning, observer frame, and photometric shock

7.1 Observer-frame compensation

The source name “Lorentz frame drift” is retained only as a historical alias. For ordinary transit, robotics, or camera motion, the formal operator is an observer-frame transform in $SE(d)$, not a relativistic Lorentz transformation.

Let $g_n = (R_n, t_n) \in SE(d)$ map sensor coordinates into a stabilized frame. A point and velocity are conditioned by

$$p_n^{\text{stab}} = R_n p_n^{\text{sensor}} + t_n, \tag{7.1}$$
$$v_n^{\text{rel}} = v_n^{\text{object}} - v_n^{\text{frame}}. \tag{7.2}$$

In a rotating three-dimensional frame, the implementation includes the standard angular-velocity terms or performs the complete homogeneous transform before differencing. The estimated frame state and its uncertainty are part of the input quality record.

7.2 Stationary calibration window

A low-motion interval can calibrate sensor bias, clock offset, background phase, and exposure. Define a stationary gate

$$q_{\text{stat}}[n] = \mathbf{1}\left[\|\widehat{v}_{\text{frame}}[n]\| \le \tau_v \ \wedge \ \|\widehat{\alpha}_{\text{frame}}[n]\| \le \tau_a\right]. \tag{7.3}$$

A calibration update is accepted only after $q_{\text{stat}} = 1$ for a declared dwell time and after residual checks pass. Stopping at a platform or workstation can therefore be a calibration opportunity, but the location itself is not a calibration certificate.

7.3 Photometric shock adapter

A sudden transition from dark to bright conditions is a sensor and photometric problem. Let $Y_n(x)$ be luminance and

$$\ell_n(x) = \log(\varepsilon_Y + Y_n(x)). \tag{7.4}$$

Define saturation fraction and shock magnitude

$$s_n = \frac{1}{|\Omega|} \sum_{x\in\Omega} \mathbf{1}[Y_n(x) \ge Y_{\max}], \tag{7.5}$$
$$q_n = \left|\text{median}(\ell_n) - \text{median}(\ell_{n-1})\right|. \tag{7.6}$$

When $s_n > \tau_s$ or $q_n > \tau_q$, the input conditioner may freeze fragile state updates, lower exposure/gain, select an HDR path, and emit a quality flag. Spatial Möbius inversion may reallocate coordinates, but it cannot remove charge overflow, blooming, clipping, or missing photons in a physical sensor.

Evidence boundary

Anti-blooming, dynamic range, rolling shutter, and exposure are properties of the sensor and acquisition pipeline. A coordinate transform or NOT gate cannot guarantee their correction. Any camera claim requires measured raw-sensor and reconstructed-image results.

7.4 Anisotropic region-of-interest operator

The commute transcript's narrow "vision slit" becomes an explicit mask

$$m_n(x, y) \in \{0, 1\}, \qquad X_n^{\mathrm{ROI}}(x, y) = m_n(x, y) X_n(x, y). \tag{7.7}$$

For a horizontal strip, $m_n = 1$ when $|y - y_f| \leq h/2$. The operator reduces work in proportion to the retained samples, but it also removes information. A conforming profile records mask geometry, dropped coverage, and the safety conditions under which the mask may be used.

8

Phase control, parity conditioning, and orientation

8.1 Second-order phase variation

After phase unwrapping or a declared wrapped-difference method,

$$\Delta^2\phi_i[n] = \mathrm{wrap}\left(\phi_i[n] - 2\phi_i[n-1] + \phi_i[n-2]\right). \tag{8.1}$$

Because second differences amplify noise, the input phase **SHOULD** be filtered and its sample rate justified. Branch angle and scale may be modulated by

$$\beta_i[n] = \beta_0 + k_\phi\,\mathrm{clip}(\Delta^2\phi_i[n], -\delta_{\max}, \delta_{\max}), \tag{8.2}$$

$$s_i[n] = \varphi_g^{-\eta}\exp\left(-k_s|\Delta^2\phi_i[n]|\right), \qquad \eta > 0. \tag{8.3}$$

The default $s_i < 1$ supports bounded recursive length.

8.2 Discrete phase-locked loop

Let ϕ_i^{meas} be measured phase and $\widehat{\phi}_i$ the predicted phase. Define

$$e_{\phi,i}[n] = \mathrm{wrap}(\phi_i^{\mathrm{meas}}[n] - \widehat{\phi}_i[n]). \tag{8.4}$$

A reference second-order digital lock is

$$\widehat{\omega}_i[n+1] = \mathrm{clip}\left(\widehat{\omega}_i[n] + k_I e_{\phi,i}[n], -\omega_{\max}, \omega_{\max}\right), \tag{8.5}$$

$$\widehat{\phi}_i[n+1] = \mathrm{wrap}\left(\widehat{\phi}_i[n] + \Delta t_n \widehat{\omega}_i[n+1] + k_P e_{\phi,i}[n]\right). \tag{8.6}$$

The lock state becomes true only when $|e_{\phi,i}| \le \tau_\phi$ for a declared dwell count. A parity event may request reacquisition; it is not itself a phase estimator.

8.3 Debounce, hysteresis, and event rate

The raw parity bit may be filtered by a causal vote

$$\bar{p}_i[n] = \mathbf{1}\left[\sum_{r=0}^{h-1} p_i[n-r] \ge m\right], \qquad 1 \le m \le h. \tag{8.7}$$

The toggle rate is

$$f_{p,i}[n] = \frac{1}{h\Delta t}\sum_{r=1}^{h} |p_i[n-r+1] - p_i[n-r]|. \tag{8.8}$$

Excess rate, saturation, unlock, or resource pressure enters safe mode rather than driving uncontrolled branch oscillation.

8.4 Orientation transition and the y-axis example

A coordinate sign changes only when a declared chart transition or application transform requires it. If $o_i \in \{+1, -1\}$ is the chart orientation, a two-dimensional projection may use

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & o_i \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \tag{8.9}$$

A parity bit can report the odd/even count of orientation reversals, but it does not arbitrarily flip a sign merely because a clock digit changes.

Interpretation of the 07:50–07:51 observation

The observed time-digit transition is useful as a human-readable example of a discrete edge and generation change. In the formal engine it motivates monotonic generation tags and event logging, not a physical proof that the right-hand bit controls a spatial axis.

9

Budgeted attention, interrupt, and recovery

9.1 Foveation as resource allocation

The commute transcript's attention language is formalized as an engineering allocation policy, not a diagnosis or model of a person. Let $d_{\mathrm{LP}}(u_i, u_f)$ be a declared distance in log-polar address space from cell i to a task-selected focus u_f. Define a saliency score

$$s_i = \lambda_q q_i^{\mathrm{task}} + \lambda_r \exp\left(-\frac{d_{\mathrm{LP}}(u_i, u_f)^2}{2\sigma_f^2}\right) + \lambda_e q_i^{\mathrm{event}}, \tag{9.1}$$

where q_i^{task}, q_i^{event}, and all weights are application-defined and logged.

The allocator chooses activity levels $b_i \in [b_{\min}, 1]$ under a finite budget:

$$\max_{b_i} \sum_i s_i b_i \quad \text{subject to} \quad \sum_i c_i b_i \le B_n, \qquad b_i \ge b_{\min} > 0 \text{ for safety-relevant peripheral cells.} \tag{9.2}$$

The nonzero floor prevents total blindness outside the focus. "Perfect attention at zero effort" is therefore not a normative claim; finite sensing and computation always have cost.

9.2 Interrupt state

A high-priority event may enter a focused mode when

$$q_i^{\mathrm{event}} > \tau_e \quad \text{or} \quad r_i^{\mathrm{safety}} < \tau_{\mathrm{near}} \quad \text{or} \quad q_n^{\mathrm{shock}} > \tau_q. \tag{9.3}$$

The interrupt handler performs four bounded actions:

1. store the current generation and active-cell digest in a ring-buffer snapshot;
2. reserve a fixed work budget for the event region;
3. reduce, but do not eliminate, background update rates; and
4. set an expiration time and revalidation condition.

No unbounded recursive branch or hidden allocation is permitted during an interrupt.

9.3 Causal coherence recovery

Let z_i^{cache} be the last valid background state and z_i^{new} the reacquired state. In Euclidean components a simple recovery filter is

$$z_i^{\mathrm{rec}}[n+1] = (1-\alpha_i) z_i^{\mathrm{cache}} + \alpha_i z_i^{\mathrm{new}}, \qquad 0 < \alpha_i \le 1. \tag{9.4}$$

For chart or manifold coordinates, interpolation uses a declared geodesic or local chart and checks seam orientation. Recovery is complete only after boundary residual, phase lock, and generation consistency pass.

9.4 Attention telemetry

A conforming run records the focus address, active budget, peripheral floor, interrupt cause, dwell time, dropped samples, recovery residual, and any safety-region coverage loss. This turns the transcript's "afterimage" or "distraction latch" into a reproducible state transition rather than an unverifiable mental-state claim.

10

Golden-ratio BST/L-system router

10.1 Grammar and ordering

Let $B(d, \ell, \psi, \kappa)$ denote a branch node at depth d, length ℓ, direction ψ, and ordering key κ. A gated bifurcation is

$$B(d, \ell, \psi, \kappa) \xrightarrow{\bar{p}=1} \begin{cases} B(d+1, s\ell, \psi + \beta, \kappa_L), \\ B(d+1, s\ell, \psi - \beta, \kappa_R), \end{cases} \tag{10.1}$$

subject to

$$\kappa_L < \kappa_P < \kappa_R. \tag{10.2}$$

The geometric L-system rule and BST ordering are distinct constraints applied to the same node records.

10.2 Bounded allocation

The active node count obeys

$$0 \le N_n \le N_{\max}. \tag{10.3}$$

A deterministic overflow policy must select among rejecting low-priority proposals, recycling expired nodes, merging nearby leaves, or reducing branch rate. Branch allocation is a separate scan/queue stage; it may not be hidden in a function that can silently overrun memory.

10.3 Golden ratio as a hypothesis

The golden ratio is preserved because it is part of the source definition, but it is not privileged by proof. A conformance profile declares exactly how φ_g affects length, angle, priority, or quantization. Validation must compare it with fixed, optimized, random, and application-specific alternatives at equal node and compute budgets.

11

Adaptive forward timeline and workload guard

11.1 Directional adherence

Every update has a strictly increasing generation g_n. A physical buffer slot stores (g, z), and a read for generation g_n rejects any slot tagged $g > g_n$. This implements the source's one-dimensional forward timeline without claiming zero latency or forbidding bounded ring-buffer wrap.

11.2 Resolution-aware time step

For a log-polar cell at radius r_i, a conservative local physical resolution is

$$\ell_i = r_i \min\big((\ln b)\Delta w, \Delta\theta\big). \tag{11.1}$$

Let V_i and A_i bound relative speed and acceleration over the next step. A resolution-aware step satisfies

$$V_i \Delta t + \frac{1}{2} A_i \Delta t^2 \leq \eta_\ell \ell_i, \qquad 0 < \eta_\ell < 1. \tag{11.2}$$

Thus

$$\Delta t_i \leq \begin{cases} \dfrac{\sqrt{V_i^2 + 2A_i \eta_\ell \ell_i} - V_i}{A_i}, & A_i > 0, \\ \dfrac{\eta_\ell \ell_i}{V_i + \varepsilon}, & A_i = 0. \end{cases} \tag{11.3}$$

The global step is the minimum of nominal, resolution, contact, sensor, and deadline bounds.

11.3 Workload governor

Let C_n estimate remaining work and D_n the deadline budget. The governor controls active cells, branch proposals, sweep samples, and output resolution so that

$$C_n \leq D_n - \Delta_{\text{reserve}}. \tag{11.4}$$

When the bound cannot be met, the engine degrades a declared noncritical output, preserves safety cells, freezes speculative branches, or enters safe mode. It does not relabel dropped computation as "time dilation."

Table 11.1: Reference runtime modes.

Mode	Trigger	Required behavior
Calibrate	sustained low frame motion and valid sensors	Estimate bias/exposure/phase baseline; do not overwrite last trusted state until checks pass.
Transit	observer motion above threshold	Enable frame compensation and resolution-aware time step.
Focused	event/saliency threshold	Reallocate bounded work while maintaining peripheral floor.
Shock	photometric or acoustic discontinuity	Freeze fragile state, adapt frontend, retain quality flags.
Contact	predicted gap below threshold	Invoke conservative advancement and collision response.
Safe	parity rate, unlock, overflow, deadline, or residual alarm	Stop new branching, damp motion, preserve last valid generation, request external handling.
Stationary	low motion after transit	Permit calibration, compaction, and provenance checkpoint.

12

General-purpose CCD contract

12.1 Why the Version 0.2 profile was not general-purpose

Version 0.2 supplied a valid conservative-advancement idea for a point or reduced configuration with a trustworthy gap function. That is sufficient for some point–implicit, sphere–implicit, and configuration-space queries, but it leaves five essential questions unanswered for a literal graphics-engine CCD subsystem:

1. Which finite shapes, meshes, compounds, or deforming features define the gap?
2. Which broad phase guarantees that no potentially colliding pair is omitted?
3. Which narrow-phase algorithm applies to each geometry and motion combination?
4. How are floating-point degeneracies, simultaneous contacts, initial overlap, and termination reported?
5. How is the first-contact result coupled to a multi-body integrator without creating new interpenetration?

Rigid-body CCD, deformable-mesh CCD, and robust interval/inclusion methods are established research areas rather than consequences of an SDF lookup alone [1, 2, 4, 14]. Version 0.3 therefore treats CCD as a suite of certified backends under one query contract.

12.2 Meaning of general-purpose

A Version 0.3 general-purpose CCD implementation supports a declared matrix of common geometry and motion classes through capability-based dispatch. At minimum, the specification provides contracts for:

- analytic primitives and bounded convex shapes;
- rigid bodies with translation and rotation;
- triangle surface meshes and optional volume elements;
- linearly moving deformable vertices and self-collision;
- compound or concave shapes through bounded decomposition or feature hierarchies;
- height fields, voxels, or implicit fields when a conservative separation/lower-bound oracle exists; and
- scripted or polynomial motion after subdivision into intervals for which the declared backend assumptions hold.

General-purpose does not mean universal. A backend may return `UNSUPPORTED` when a capability is absent and `INDETERMINATE` when its numerical or geometric assumptions fail. Exposed incompleteness is part of conformance.

12.3 Query record

A pair query is

$$Q = (A, B, \mathcal{M}_A, \mathcal{M}_B, [t_0, t_1], d_{\min}, \tau_x, \tau_t, W, \pi), \tag{12.1}$$

where A, B are shape proxies; $\mathcal{M}_A, \mathcal{M}_B$ motion models; $[t_0, t_1]$ the query interval; $d_{\min} \geq 0$ a requested minimal separation or thickness; τ_x and τ_t spatial and temporal tolerances; W the work/iteration budget; and π a policy containing certification, determinism, response, and fallback requirements.

The query interval is closed. The first contact time is

$$t^* = \inf\{t \in [t_0, t_1] : d(A(t), B(t)) \le d_{\min}\}, \tag{12.2}$$

provided the separation function and motion are defined throughout the interval. If the pair begins within $d_{\min}$, the result is an initial-overlap/contact state rather than a future TOI.

12.4 Result status and certificate class

Every result carries one of the following normative statuses:

Status	Meaning
`CLEAR_CERTIFIED`	The backend certifies that separation exceeds $d_{\min}$ throughout the interval.
`HIT_CERTIFIED`	A first-contact interval $[\underline{t}, \overline{t}]$ encloses t^*, with terminal width at most τ_t or a stronger exact predicate.
`HIT_CONSERVATIVE`	Contact is reported no later than the true first contact, but the interval may be wider or contain a false positive under the declared conservative approximation.
`INITIAL_CONTACT`	The pair starts at or within $d_{\min}$; no positive future TOI is claimed.
`INITIAL_OVERLAP`	The pair starts in penetration beyond the accepted overlap tolerance; a separate recovery policy is required.
`INDETERMINATE`	The method exhausted work, encountered unresolved degeneracy, or lost a required bound; no clear result is certified.
`UNSUPPORTED`	The geometry/motion/certification combination is outside the registered backend capabilities.
`NUMERIC_FAILURE`	Non-finite arithmetic, invalid interval, corrupt geometry, or inconsistent witness data was detected.

The certificate class γ is one of the following: `EXACT_PREDICATE`, `INCLUSION`, `CONSERVATIVE_ADVANCEMENT`, `ANALYTIC_BOUNDED`, or `APPROXIMATE`. Only the first four may produce a certified no-tunneling result. An approximate or speculative contact backend may exist as an application profile but may not be relabelled as certified CCD.

12.5 Minimal separation and scale

A positive $d_{\min}$ is often preferable to exact zero because it can represent physical thickness and can provide a numerical safety envelope. It is not a hidden tolerance: its units and derivation **MUST** be declared. The total spatial allowance is

$$d_{\text{guard}} = d_{\min} + \varepsilon_{\text{geom}} + \varepsilon_{\text{motion}} + \varepsilon_{\text{arith}}, \tag{12.3}$$

where each error term is a conservative bound or is zero. Scene scaling must be handled explicitly; one fixed epsilon for all coordinate magnitudes is non-conforming.

12.6 Initial overlap is a separate problem

CCD answers whether a previously valid separated configuration reaches contact. If the start state is already overlapping, the engine must not invent a negative TOI. It returns `INITIAL_OVERLAP` with penetration witnesses when available and invokes a declared recovery method: rollback to the last valid generation, penetration-depth/MTD projection, a barrier line search, or fail-safe freeze. Recovery is allowed to be approximate, but its uncertainty remains visible.

12.7 Core no-crossing obligation

For every result labelled certified, the engine must establish one of the following:

1. a proof that no root/contact exists on the interval;
2. an interval containing the earliest root/contact;
3. a sequence of conservative advancement steps that cannot pass contact under their declared bounds; or
4. an exact predicate coupled to a conservative time-localization rule.

The benchmark literature documents that fast algorithms can produce false negatives and that conservative algorithms can produce false positives; Version 0.3 therefore measures both and forbids a certified false negative [14, 16].

13

Motion, geometry, and separation contracts

13.1 Motion capability record

Each object supplies a motion record over $I = [t_0, t_1]$:

$$\mathcal{M} = (X(t), \dot{X} \text{ or bounds}, \mathcal{E}_X(I), c_X), \tag{13.1}$$

where $X(t)$ is the pose or vertex map, $\mathcal{E}_X(I)$ is a conservative trajectory enclosure, and c_X names the capability class.

Normative capability classes are:

LINEAR_TRANSLATION $p(t) = p_0 + u(t)(p_1 - p_0)$ with $u = (t - t_0)/(t_1 - t_0)$.

RIGID_CURVED $x(t) = R(t)x_0 + p(t)$ with declared interpolation and derivative/angle bounds.

VERTEX_LINEAR Every mesh vertex follows a linear trajectory over the current interval.

POLYNOMIAL_k Coordinates are polynomial or spline segments with a conservative interval evaluator.

ORACLE_BOUNDED The motion is evaluated by an oracle that supplies conservative position, speed, and acceleration bounds.

A discontinuity in pose, topology, teleport, scripted state, or boundary condition splits the timeline. No CCD query may span an undeclared discontinuity.

13.2 Rigid translation, rotation, and acceleration bounds

For bodies A and B with reference-point translational velocities v_A, v_B, angular speed bounds ω_A, ω_B, and bounding radii R_A, R_B about their rotation centers, a conservative point-speed bound is

$$V_{AB} \le \|v_A - v_B\| + |\omega_A| R_A + |\omega_B| R_B. \tag{13.2}$$

This bound is deliberately geometry-wide; local witness bounds may be tighter. With relative acceleration bound A_{AB}, the maximum possible closing distance over $\tau \ge 0$ is

$$\Delta d_{\text{close}}(\tau) \le V_{AB}\tau + \frac{1}{2} A_{AB}\tau^2. \tag{13.3}$$

Given current lower-bound gap $g > d_{\text{guard}}$, a conservative accelerated step is the nonnegative root of

$$\frac{1}{2} A_{AB}\tau^2 + V_{AB}\tau - (g - d_{\text{guard}}) = 0, \tag{13.4}$$

multiplied by a safety factor $\eta \in (0, 1)$. When $A_{AB} = 0$, the expression reduces to gap divided by speed.

Curved rigid trajectories may also be subdivided into linearized subintervals whose geometric deviation is bounded and absorbed into d_{guard}. Rigid-IPC demonstrates one such strategy: conservative curved-motion queries are reduced to linear CCD queries with minimal separation [19].

13.3 Shape proxy capabilities

A shape proxy is a finite typed interface rather than an assumption that every object has one exact global SDF. It may expose:

- a conservative axis-aligned or oriented bound over a time interval;
- a support map $s_A(d) = \arg\max_{x \in A} d \cdot x$ for convex queries;
- a static or time-dependent distance/lower-bound oracle and witness data;
- primitive parameters for analytic solvers;
- vertices, edges, faces, adjacency, and feature identifiers;
- a BVH or volume-element hierarchy;
- an implicit function $F(x, t)$ with spatial Lipschitz, temporal-rate, gradient, and approximation-error bounds;
- decomposition children for compounds/concave shapes; and
- generation tags that invalidate cached pairs when geometry or topology changes.

A backend advertises the capabilities it consumes. Dispatch succeeds only when both query and backend contracts match.

13.4 Separation oracle

A separation oracle at time t returns

$$\mathcal{O}(t) = (\underline{d}(t), x_A(t), x_B(t), n(t), B_d(t), \varepsilon_d, \chi), \tag{13.5}$$

where $\underline{d}$ is a conservative lower bound on true separation, x_A, x_B are witnesses when defined, n is a separating/contact normal, B_d bounds closing rate, ε_d is oracle error already accounted for in the lower bound, and χ records validity/degeneracy.

The lower-bound obligation is

$$\underline{d}(t) \le d(A(t), B(t)). \tag{13.6}$$

Overestimating separation can cause tunneling and is forbidden for certified advancement. Underestimating separation is conservative but may increase false positives or work.

13.5 Implicit and SDF contract

For an implicit field $F(x, t)$ representing a static obstacle boundary, a certified point or sphere query requires a declared relationship between F and Euclidean separation. One valid contract is

$$d(x, \partial\Omega_t) \ge \frac{\max(F(x, t) - \varepsilon_F, 0)}{L_F}, \tag{13.7}$$

where L_F is a spatial Lipschitz bound and ε_F bounds field approximation error. If F is an exact signed distance outside the obstacle, $L_F = 1$ and $\varepsilon_F = 0$ away from singularities.

For a moving query point with speed bound V and temporal field-rate bound $\Lambda_t \ge |\partial_t F|$, a safe step is

$$\Delta t \le \eta \frac{\max(F - \varepsilon_F - L_F d_{\text{guard}}, 0)}{L_F V + \Lambda_t + \varepsilon_r}. \tag{13.8}$$

A sampled texture called an SDF is not automatically valid. The construction, interpolation, truncation, reinitialization, gradient, and sampling errors must be bounded or the result is approximate.

13.6 Configuration-space interpretation

For convex sets under pure translation, collision between $A(t)$ and $B(t)$ is equivalent to the relative translation ray entering the Minkowski difference $A \ominus B$. Support-mapped shape casting can therefore use a GJK-style distance or ray-cast iteration. Under rotation, the configuration-space obstacle changes with time, so a pure translational shape cast is insufficient; the backend must use angular bounds, direct rigid-motion mathematics, or conservative subdivision.

13.7 Transforms and local coordinates

To reduce arithmetic range, a backend may transform both objects into a common local frame. The transform must be invertible on the query interval and must conservatively transform $d_{\min}$, error bounds, velocities, and witness normals. Nonuniform scaling changes distance and normal metrics; it cannot be treated as a free coordinate rename.

14

Conservative broad phase and pair generation

14.1 Candidate completeness invariant

Let $\mathcal{P}^*$ be the set of object or feature pairs whose swept geometry reaches separation d_{guard} anywhere in $[t_0, t_1]$. A certified broad phase produces candidates $\mathcal{P}$ satisfying

$$\mathcal{P}^* \subseteq \mathcal{P}. \tag{14.1}$$

False positives are permitted; false negatives invalidate every downstream CCD certificate. A broad phase that is fast but approximate may be used only in a non-certified profile or behind a complete fallback.

14.2 Swept bounds

For shape A with trajectory enclosure $\mathcal{E}_A(I)$, its inflated swept AABB is

$$B_A(I) = \text{AABB}(\mathcal{E}_A(I)) \oplus [-d_{\text{guard}}, d_{\text{guard}}]^3. \tag{14.2}$$

A pair is rejected only when $B_A(I) \cap B_B(I) = \varnothing$. For linear vertex motion, taking componentwise minima and maxima over both endpoints is exact for the AABB of each vertex trajectory. For curved rigid motion, the enclosure must include rotational arcs, for example by endpoint bounds plus a certified angular-radius expansion or by interval subdivision.

14.3 Hierarchies and temporal coherence

Complex models use BVHs such as AABB, OBB, or k-DOP hierarchies. Hierarchical culling is established for complex rigid and deformable geometry [6, 4]. Version 0.3 requires:

- each internal node bound encloses every descendant throughout the interval;
- refit or rebuild occurs after deformation/topology changes according to a declared quality criterion;
- cached front/pair lists carry geometry generations and are invalidated when stale;
- traversal capacity is finite and overflow cannot silently drop pairs; and
- the root bounds are post-checked for finite values and ordering.

14.4 Sweep-and-prune and spatial hashing

Sweep-and-prune, grids, and spatial hashes are valid when their swept intervals or occupied cells are conservative. Quantizing a bound to cells must round outward. A hash collision may add work but may not remove a pair. The scalable CCD literature provides a broad/narrow phase benchmark and parallel algorithms designed around conservative floating-point behavior [16].

14.5 Self-collision pair policy

For a mesh, adjacent features sharing a vertex or edge often represent legal local connectivity rather than collision. The self-collision policy declares excluded topological neighborhoods, duplicate feature-pair canonicalization, and any normal-cone or connectivity culling. Such culling must be justified by geometry/topology invariants; it may not exclude pairs merely because they were not colliding in the previous frame. Connectivity-based culling can greatly reduce deformable CCD work but remains part of the completeness proof [4].

14.6 Pair identity, deduplication, and capacity

A canonical pair key is

$$k_{ij} = (\min(i, j), \max(i, j), g_i, g_j, f_i, f_j), \tag{14.3}$$

including object/geometry generations. Candidate queues are deduplicated deterministically before narrow phase when strict replay is required. Queue overflow invokes one of three declared policies: spill to a larger bounded queue, divide the time/space interval and retry, or return a visible resource failure. Dropping pairs is forbidden.

14.7 Broad-phase audit

A debug/certification mode randomly or exhaustively samples pairs on small scenes and compares the candidate set to brute-force swept-bound overlap. The following counters are mandatory: total objects/features, hierarchy nodes visited, candidate pairs before/after deduplication, overflow count, stale-generation rejection, and broad-phase certificate status.

15

Narrow-phase backend suite

15.1 Capability-based dispatch

No single numerical method is optimal for all geometry and motion. The dispatcher selects the narrowest backend whose declared capability dominates the query. A representative matrix is shown below.

Table 15.1: Reference narrow-phase dispatch matrix.

Geometry pair	**Motion**	**Preferred backend**	**Certificate basis**	**Fallback**
Sphere–sphere, point–plane, capsule primitives	Linear or bounded polynomial	Analytic polynomial TOI	Outward-rounded roots and interval verification	Interval subdivision
AABB–AABB	Linear translation	Slab time-interval intersection	Exact interval inequalities in declared arithmetic	Swept-bound conservative hit
Convex–convex	Pure translation	Support-map shape cast/GJK ray cast	Separating distance and monotone relative ray	Inclusion/CA
Convex–convex	Rigid curved motion	Conservative advancement with angular bound	Lower-bound gap divided by certified closing rate	Curved-motion subdivision
Convex–triangle/mesh	Rigid or linearized	BVH traversal plus convex/-triangle backend	Complete hierarchy and child certificate	Triangle feature CCD
Triangle–triangle surface mesh	Vertex-linear	Vertex–face and edge–edge CCD	Polynomial/inclusion/exact-predicate feature tests	Tight inclusion subdivision
Volume elements	Vertex-linear/deforming	Continuous SAT plus surface/feature tests	Conservative separating axes and feature completeness	Surface extraction plus CCD
Point/sphere–implicit	Bounded trajectory	Lipschitz/error-aware conservative advancement	Certified field lower bound and rate bound	Interval field inclusion
Compound/concave	Supported children	Child swept hierarchy and earliest child TOI	Complete decomposition/hierarchy	Conservative union hit
Heightfield/voxel	Linear or bounded	Cell DDA/swept hierarchy plus primitive backend	Conservative cell occupancy/-field bound	Dense interval search
Arbitrary black-box	Any	None	None	`UNSUPPORTED`

15.2 Analytic primitive solvers

When relative motion and primitive equations produce a low-degree polynomial, a backend should use the analytic structure but verify candidate roots against the original geometry. For spheres with relative center $r(t) = r_0 + vt$ and combined radius R, contact solves

$$\|r_0 + vt\|^2 - R^2 = 0. \tag{15.1}$$

The solver handles initial contact, zero relative velocity, negative discriminant, tangency, and interval clipping. Discriminants and roots are evaluated with outward error bounds or verified by interval substitution before a certified status is returned.

The same principle applies to point–plane, moving interval/AABB slabs, and selected capsule/segment cases. Analytic does not mean numerically exact; verification remains mandatory.

15.3 Convex translational shape casting

For convex bodies under relative translation, a support-mapped ray cast in the Minkowski difference can compute earliest contact. Each iteration obtains a separating direction and advances along the relative ray until the origin reaches the inflated Minkowski set. A conforming implementation records support-map validity, simplex degeneracy, progress, and an iteration limit. Stagnation returns `INDETERMINATE` or switches to an inclusion fallback; it does not return clear.

15.4 Rigid convex conservative advancement

For general rigid motion, let g_k be a conservative separation lower bound at t_k and B_k a bound on closing rate that includes translation and rotation. The step is

$$t_{k+1} = t_k + \eta \frac{\max(g_k - d_{\text{guard}}, 0)}{B_k + \varepsilon_r}. \tag{15.2}$$

Controlled conservative advancement and related rigid-body CCD methods use distance bounds and hierarchical culling to approach first contact without crossing it [1, 2]. Version 0.3 additionally requires a progress floor, upper time clamp, witness validation, angular/acceleration accounting, and a certified fallback near degeneracy.

15.5 Triangle-mesh elementary tests

For linearly moving triangle meshes, first contact between two non-intersecting triangles can be reduced to vertex–face (VF) and edge–edge (EE) feature events. A VF coplanarity function is

$$f_{\text{VF}}(t) = \big(p(t) - a(t)\big) \cdot \Big(\big(b(t) - a(t)\big) \times \big(c(t) - a(t)\big)\Big). \tag{15.3}$$

With linear trajectories, f_{VF} is at most cubic. A root is a collision only when the point lies in the triangle at that time and the separation/thickness condition is satisfied. EE uses an analogous coplanarity equation plus segment parameters.

Direct cubic solving is fragile near planar and degenerate cases. Geometrically exact predicates, Bernstein sign classification, and interval/inclusion methods have been developed specifically to avoid missed mesh contacts [11, 12, 14]. The Version 0.3 certified reference profile prefers inclusion subdivision with filtered/exact terminal predicates; a faster polynomial solver may be registered only with an error defence and post-check.

15.6 Deforming surface and volume meshes

For deformable surfaces, broad-phase hierarchy refit and connectivity-aware culling feed VF/EE tests. For volume elements, continuous separating-axis culling can reject element pairs before testing relevant features; this has been demonstrated for deforming volume meshes, including topology changes [5]. A backend must define whether it detects only boundary contact, interior element overlap, inversion, or all three.

15.7 Implicit-field backend

The implicit backend uses the contract in chapter 13. It is well suited to static or smoothly moving analytic fields, point/sphere queries, and configuration-space fields with certified distance bounds. It is not automatically suitable to arbitrary triangle soups, discontinuous CSG, under-resolved textures, or a globally zero NHDF field. If the gradient is singular, the field lower bound is invalid, or the rate bound becomes non-finite, the backend brackets/subdivides or returns indeterminate.

15.8 Compounds and concave objects

A compound or concave object is handled by a finite hierarchy of supported children, a conservative convex decomposition, or a mesh backend. The object-pair TOI is the minimum over all child-pair TOIs that survive broad phase. Child pruning must be conservative, and child transforms/generations are part of the pair key. A concave object may not be passed to a convex support-map backend unless the supplied support map truly represents the convex hull and the resulting hull conservatism is accepted.

15.9 Backend registration rule

Every backend registration contains:

$$(G_A, G_B, M_A, M_B, \gamma, d_{\min}\text{-support}, \mathcal{A}, \mathcal{F}, \mathcal{V}), \tag{15.4}$$

where G are geometry classes, M motion classes, γ certificate class, $\mathcal{A}$ assumptions, $\mathcal{F}$ failure statuses, and $\mathcal{V}$ verification vectors. Dispatch is deterministic for a given policy and registry version.

16

Certified time-of-impact localization and numerical robustness

16.1 Time interval, not just a float

The primary output is an interval $[\underline{t}, \bar{t}]$ containing the earliest contact. A single floating-point TOI may be supplied as a convenience value, but the certificate references the interval and its construction. For conservative response, the engine advances to a protected time

$$t_{\text{safe}} = \max(t_0, \underline{t} - \tau_{\text{lead}}), \tag{16.1}$$

where τ_{lead} is a declared nonnegative lead allowance. The interval must never be shifted past a certified first contact.

16.2 Inclusion subdivision

Let $F(Q, I)$ be an interval extension whose range encloses the contact equations for query Q over time interval I. The generic inclusion algorithm is:

1. reject I when $0 \notin F(Q, I)$ or when a conservative separation lower bound exceeds d_{guard};
2. accept or terminal-test I when its width is at most τ_t or an exact predicate resolves it;
3. otherwise split I according to a deterministic rule and visit the earlier child first;
4. stop at the earliest accepted interval; and
5. return indeterminate if work bounds are exhausted before a certificate is obtained.

Interval analysis has long been used for time-dependent interference, and modern inclusion CCD combines it with robust predicates and explicit false-positive/runtime tradeoffs [9, 10, 14].

16.3 Reference conservative-advancement loop

For a separation oracle $\mathcal{O}(t)$, the reference loop is:

Listing 16.1: Certificate-carrying conservative advancement.

```
t = t0
last_clear = t0
for k in 0 .. max_iterations-1:
    sample = oracle(t)
    require finite(sample.lower_gap, sample.closing_rate_bound)
    if sample.lower_gap <= guard_distance:
        return localize_in_bracket(last_clear, t, query)
    if sample.closing_rate_bound <= rate_floor:
        return CLEAR_CERTIFIED if static_separation_is_certified else INDETERMINATE
    dt = safety * (sample.lower_gap - guard_distance) /
         sample.closing_rate_bound
    dt = clamp(dt, min_progress, t1 - t)
    if dt <= 0:
        return INDETERMINATE
    last_clear = t
    t = t + dt
    if t >= t1:
        return CLEAR_CERTIFIED
return INDETERMINATE
```

The actual implementation may use acceleration roots, local Lipschitz constants, or hierarchical bounds. Each optimization preserves the lower-gap and closing-rate obligations.

16.4 Filtered and exact predicates

A filtered predicate first evaluates in ordinary floating point with a rigorously computed error bound. If the result is separated from zero by more than that bound, its sign is certified. Otherwise it falls back to wider precision, interval arithmetic, expansions, rational arithmetic, or a geometrically exact construction. Exact Boolean predicates can be combined with conservative temporal bisection; they do not by themselves produce the earliest time.

Root-parity methods can exactly count parity for a cubic CCD predicate, but parity alone cannot distinguish no roots from some even-root cases; it is therefore a component, not a universal TOI certificate [15]. Version 0.3 uses root parity only when an additional rule resolves that ambiguity.

16.5 Degenerate configurations

The terminal kernel must classify at least:

- zero-area or nearly zero-area triangles;
- zero-length or nearly zero-length edges;
- parallel or nearly parallel edge pairs;
- coplanarity over a time interval rather than at an isolated root;
- tangency/double roots;
- coincident features and duplicate topology;
- non-finite coordinates and extreme scaling; and
- start/end contact exactly at t_0 or t_1.

A specialized dimension-reduced test may be used. Otherwise the backend inflates to d_{guard} and invokes interval distance inclusion. A degeneracy flag is part of the result.

16.6 Rounding and tolerance policy

Tolerances are derived from units, coordinate scale, representation precision, and error analysis. The policy includes:

$$\tau_x = \tau_{\mathrm{abs}} + \tau_{\mathrm{rel}}\, S_x, \tag{16.2}$$

$$\tau_t = \tau_{t,\mathrm{abs}} + \tau_{t,\mathrm{rel}}(t_1 - t_0), \tag{16.3}$$

where S_x is a declared local scale such as the maximum coordinate norm, shape radius, or BVH extent. Outward rounding is used for swept bounds and intervals. Error-defence methods are required for polynomial/root implementations that otherwise rely on tuned epsilons [13].

16.7 Post-check and rollback

After advancing to a certified safe time, the engine executes a post-check using an independent or stronger predicate whenever feasible. If the check detects forbidden overlap, it rolls back to the last certified generation, halves or otherwise reduces the interval, invalidates the responsible cache/backend certificate, and records a failure. A post-check is defence in depth; it does not excuse an incomplete broad phase.

16.8 Termination and progress

Every iterative backend declares maximum iterations, maximum interval subdivisions, minimum progress, and a work counter. To avoid falsely reporting clear when the bound is exhausted, exhaustion returns indeterminate. The application policy then chooses among smaller global step, stronger backend, CPU/exact fallback, or fail-safe stop.

16.9 Deterministic replay

A strict deterministic profile fixes pair ordering, backend registry version, interval split direction, tie-breaks, reduction order, rounding mode where controllable, and contact clustering. Parallel execution may reorder independent work internally, but the final result is sorted and reduced by stable keys. Bitwise identity across heterogeneous hardware is not assumed unless measured; semantic equivalence within declared intervals is the default target.

17

Contact manifolds, simultaneous events, and self-collision

17.1 Contact record

A localized contact record is

$$C = (t_I, i, j, f_i, f_j, x_i, x_j, n, d, \dot{d}, \gamma, \chi), \tag{17.1}$$

where t_I is the TOI interval, feature IDs remain stable for the geometry generation, x_i, x_j are witnesses, n points according to a declared object-order convention, d is signed/unsigned separation under that convention, and $\dot{d}$ is relative normal speed when defined.

Normals derived from witness differences are preferred for generic distance queries. A field gradient may be used only when its orientation and conditioning are valid. At a non-smooth feature, multiple admissible normals may exist; the manifold stores a cone or a selected feature normal plus flags.

17.2 Co-time bucket

Computing one global minimum and responding immediately can make simultaneous contacts order-dependent. Version 0.3 groups contacts whose certified intervals overlap the earliest event window:

$$\mathcal{C}_0 = \{C_k : \underline{t}_k \le \bar{t}_{\min} + \tau_{\mathrm{co}}\}, \tag{17.2}$$

where $\bar{t}_{\min}$ is the upper end of the earliest accepted interval and τ_{co} is declared. The engine advances once to a safe common time and builds contact islands before response. Robust simultaneous-impact handling is a distinct problem from pairwise detection [8].

17.3 Manifold construction

Contacts are clustered by object pair, compatible normal, proximity, and feature adjacency. Redundant points are reduced while preserving torque span and boundary coverage. A rigid pair commonly keeps up to four representative points; deformable/contact-potential profiles may retain many constraints. The cap and reduction rule are explicit and deterministic.

17.4 Persistent contact

A persistent manifold cache stores feature keys, local-space anchors, normal history, accumulated impulses or friction state, and geometry generations. It accelerates resting contact but never suppresses new CCD. A cached contact is revalidated by distance and feature existence; stale contacts are discarded.

17.5 Grazing and tangency

At a tangent event, normal relative speed may be near zero even though a geometric root exists. The result is classified as grazing and retained if the minimal-separation predicate is met. The response policy may apply no restitution, maintain a contact constraint, or continue if separation is increasing and a post-check certifies clearance. A sign-change test alone is insufficient because tangency need not change sign.

17.6 Self-collision

Self-collision uses the same TOI and manifold contracts, plus:

- topological adjacency exclusions;
- duplicate pair elimination;
- material-side or normal-cone tests when justified;
- feature ownership for shared vertices/edges;
- strain or thickness policy; and
- optional collision groups to exclude intentionally joined surfaces.

A single missed self-collision can invalidate a cloth or deformable simulation, motivating accurate feature-level tests and conservative culling [7, 4].

17.7 Topology change

Fracture, cutting, remeshing, welding, or branch-generated fragments create a new geometry generation. All BVH nodes, feature IDs, cached pairs, manifolds, and warm-start data touching the changed region are rebuilt or invalidated before the next certified query. A topology event occurring inside a time step divides the interval: CCD is solved up to the event, topology is committed, then a new query begins.

17.8 Contact islands

The contact graph has dynamic bodies as vertices and active constraints as edges. Connected components are solved as islands. Static/kinematic bodies may connect otherwise separate dynamic bodies and therefore belong to island construction as anchors. Island capacity is bounded; overflow triggers partition with a declared approximation or safe failure, never silent constraint loss.

17.9 Zeno and event accumulation

Repeated impacts can create an unbounded event sequence in finite simulated time. The engine limits per-step events and transitions to a resting-contact mode when normal speed and separation fall below thresholds. Restitution is suppressed below a bounce threshold, and the remaining interval is solved by a constraint/barrier method. Exceeding event capacity without such a transition returns a visible failure.

18

Integrator coupling and collision response

18.1 Detection and response are separate contracts

CCD certifies when contact occurs. Response determines what the simulation does with that event. An accurate TOI can be paired with an inaccurate response, and a plausible response cannot repair a missed TOI. The result record and certificate remain immutable when passed into response.

18.2 Event-driven step

A reference step from T_n to $T_n + \Delta t$ is:

1. predict unconstrained trajectories for the remaining interval;
2. run broad and narrow phase;
3. if clear, commit the remaining interval;
4. otherwise advance to the certified safe event time;
5. build the co-time manifold and contact islands;
6. solve response/contact constraints;
7. reduce the remaining interval and repeat; and
8. transition to persistent/resting contact or fail when the event budget is reached.

All body forces, kinematic boundaries, joints, and animation drivers must be evaluated consistently at substep time. Reusing end-of-frame forces without declaration can break trajectory bounds.

18.3 Single rigid contact impulse

For bodies A, B, contact point offsets $r_A = x - p_A$ and $r_B = x - p_B$, inverse masses m_A^{-1}, m_B^{-1}, inverse inertia tensors I_A^{-1}, I_B^{-1} in world coordinates, and contact normal n, the relative contact velocity is

$$v_c = (v_A + \omega_A \times r_A) - (v_B + \omega_B \times r_B). \tag{18.1}$$

The normal effective inverse mass is

$$\begin{aligned} k_n = {} & m_A^{-1} + m_B^{-1} && (18.2)\\ & + n \cdot \left(\left(I_A^{-1}(r_A \times n) \right) \times r_A + \left(I_B^{-1}(r_B \times n) \right) \times r_B \right). && (18.3) \end{aligned}$$

For closing normal speed $v_n = v_c \cdot n < 0$, restitution e and optional bias $b \geq 0$ give the unclamped impulse

$$j_n = \frac{-(1+e)v_n + b}{k_n + \varepsilon_k}, \qquad j_n \geq 0. \tag{18.4}$$

Linear and angular velocities are updated by equal and opposite impulses. Restitution is disabled or reduced below a velocity threshold to avoid artificial micro-bounce.

18.4 Friction

Choose orthonormal tangents t_1, t_2. Tangential impulses are solved using their effective masses and clamped to a Coulomb disk/cone

$$\sqrt{j_{t_1}^2 + j_{t_2}^2} \le \mu j_n. \tag{18.5}$$

Static and dynamic friction may use different coefficients and a regularized transition. The basis and accumulation rule are deterministic. Merely masking velocity bits by a normal is not a physical tangent projection.

18.5 Multiple contacts and joints

A manifold/island produces a complementarity or constrained optimization problem. A real-time profile may use bounded projected Gauss–Seidel, sequential impulses, or XPBD-style constraints; an accuracy profile may use an LCP/NCP, interior-point, or barrier formulation. The solver declares iteration/tolerance limits, warm-start behavior, and residual telemetry. Joint constraints and contacts are solved in one island or with a declared splitting error.

18.6 Barrier/intersection-free profile

Incremental Potential Contact (IPC) couples CCD-filtered line search with barrier contact energies to maintain intersection- and inversion-free trajectories for deformables, and Rigid-IPC extends the concept to curved rigid-body motion and friction [17, 19]. Version 0.3 permits an IPC-like profile when its energy, distance, line-search, and solver contracts are implemented. The NHDF operators do not replace those mathematical guarantees; they may schedule, encode, or log them.

18.7 Position stabilization and overlap recovery

Small residual penetration may be corrected by a bounded projection or split impulse that does not inject excessive kinetic energy. The maximum correction, compliance, and failure threshold are explicit. Large or invalid initial overlap invokes rollback/MTD/barrier recovery rather than an arbitrarily large impulse.

18.8 Moving and kinematic boundaries

Boundary velocity at the witness point, including angular motion, participates in relative velocity and closing-rate bounds. Scripted boundary motion is either included in the CCD trajectory or applied as an equality-constrained incremental motion; teleporting a boundary after detection is non-conforming. IPC technical work specifically treats moving collision objects and boundary conditions with feasible CCD-limited steps [18].

18.9 Sleeping, resting, and reactivation

A sleeping body remains in broad-phase bounds and is reactivated by predicted contact, support loss, kinematic-neighbor motion, or external force. Resting contacts are validated each step. Sleeping may reduce work but may not remove a pair that can reach contact in the current interval.

18.10 Response failure policy

If response fails to converge within its bound, the engine does not commit a penetrating state. It rolls back to the last certified state and chooses a smaller step, stronger solver, compliant fallback, or safe freeze. Telemetry records detection certificate, response residual, and whether the state was committed.

19

Parallel and GPU general-purpose CCD profile

19.1 Stage queues

The parallel runtime uses bounded queues corresponding to the formal suboperators:

1. object/feature swept-bound generation;
2. broad-phase candidate generation;
3. canonical pair sorting and deduplication;
4. capability dispatch and backend bucketization;
5. narrow-phase iterations or interval nodes;
6. earliest-event reduction;
7. manifold/contact-island assembly;
8. response solve; and
9. certificate/telemetry emission.

Each queue has capacity, generation, overflow counter, and a policy. A queue may spill to host or a secondary pool, subdivide work, or fail visibly. It may not overwrite unprocessed entries.

19.2 Backend batching and divergence

Pairs are sorted by backend, geometry class, motion class, and optionally expected work. This reduces branch divergence and permits specialized kernels for analytic primitives, convex CA, VF, EE, or implicit queries. Persistent threads or work stealing may balance long interval subdivisions. Work ordering does not alter certificate semantics.

19.3 Parallel broad phase

A GPU profile may use parallel sweep-and-prune, LBVH, spatial hashing, or a hybrid. Bounds are rounded outward and candidate completeness is verified on a sampled audit stream. The scalable CCD project reports a parallel broad/narrow architecture and exact-ground-truth datasets intended for correctness and performance evaluation [16].

19.4 Earliest-event reduction

Each pair produces a lower TOI bound or $+\infty$. Reduction uses a stable pair key to break ties and returns an interval, not only a minimum float. Co-time collection then admits every contact whose interval overlaps the event window. Atomics that race to a single float without preserving pair/certificate data are insufficient.

19.5 Memory layout

A structure-of-arrays layout stores transforms, velocities, angular bounds, shape headers, BVH nodes, feature geometry, candidate keys, backend IDs, TOI intervals, witnesses, and status flags. Large mesh vertices and BVHs are separated from small object headers. Scratch memory is budgeted per backend and per active interval subdivision.

19.6 Determinism profile

Strict replay uses stable radix sort, deterministic scans, fixed interval split order, reproducible reductions, and a deterministic manifold/contact solver. A throughput profile may relax bitwise ordering but still must reproduce the same certificate class and enclose the reference TOI within tolerance. Cross-device differences are measured with identical test vectors.

19.7 Role of the NHDF log-polar LUT

The LUT may prioritize pairs by predicted angular/radial motion, allocate foveated contact work, cache field samples, or compress telemetry. It does not replace swept-bound completeness, feature equations, support maps, interval arithmetic, or response constraints. Any quantization error introduced by the LUT is added to d_{guard} or causes the backend to lose certified status.

19.8 Hardware claims

A conforming implementation uses only programming interfaces whose semantics are documented. Integer population count may accelerate parity, matrix units may batch transforms, and ray-query hardware may assist BVH traversal, but none automatically implements general-purpose CCD. Performance is an empirical profile, not part of the certificate.

19.9 Mandatory telemetry

Per step, the GPU profile records at least

$$\{N_o, N_f, N_p, N_d, N_q, N_{\text{hit}}, N_{\text{indet}}, N_{\text{over}}, k_{\max}, w_{\text{TOI}}, t_{\text{broad}}, t_{\text{narrow}}, t_{\text{solve}}, M_{\text{peak}}\}, \tag{19.1}$$

representing object/feature/pair counts, dispatched queries, hits, indeterminate results, overflows, maximum iterations, maximum terminal TOI width, stage times, and peak memory. Certified mode additionally records backend and certificate histograms.

20

Analytic sweeps, projections, and multimodal wavefronts

20.1 Cone and swept field

For apex a_i, unit axis u_i, and half-angle χ_i, an analytic cone surface is

$$C_i(x) = \big((x - a_i) \cdot u_i\big)^2 - \cos^2(\chi_i)\|x - a_i\|^2 = 0, \qquad (x - a_i) \cdot u_i \geq 0. \tag{20.1}$$

This is an implicit cone equation, not automatically an exact SDF. A swept field over a horizon H is

$$G_n(x) = \min_{i \in I_n} \min_{\tau \in [T_n - H, T_n]} \Psi\big(C_i(x; \tau)\big), \tag{20.2}$$

where Ψ converts residual to occupancy, distance estimate, intensity, phase delay, or another declared observable.

20.2 Projection vocabulary

Orthographic maps include

$$\Pi_{xy}(x, y, z) = (x, y), \qquad \Pi_{yz}(x, y, z) = (y, z), \qquad \Pi_{xz}(x, y, z) = (x, z). \tag{20.3}$$

A sphere projects to a circle; an aligned finite cone or pyramid has a triangular side envelope; circular cross-sections yield circles. These are calibration targets, not guaranteed output shapes for every state.

20.3 Acoustic wavefront adapter

For a source at $s(t)$ in a homogeneous medium with sound speed c, propagation delay is

$$\tau(x, t) = \frac{\|x - s(t)\|}{c}. \tag{20.4}$$

Equal-delay surfaces are spheres, or circles in a planar slice. This delay field may be used as an implicit wavefront, but acoustic pressure itself is oscillatory and is not a signed distance.

For microphone m and frequency bin f, let

$$X_m(f, n) = A_m(f, n)e^{i\phi_m(f,n)}. \tag{20.5}$$

Magnitude may be log encoded, and inter-microphone phase differences or time differences estimate direction. Second phase differences indicate frequency-rate or chirp-like behavior after unwrapping. They do not yield an exact source velocity without a source-frequency model and geometry.

20.4 Doppler relation

For collinear motion in a stationary medium, a conventional sign convention gives

$$f_{\text{obs}} = f_0 \frac{c + v_o}{c - v_s}, \tag{20.6}$$

where v_o is observer velocity toward the source and v_s is source velocity toward the observer. A siren pass can therefore serve as an acoustic calibration or event stream when f_0, clock synchronization, multipath, and uncertainty are handled. It is not proof of the engine's correctness.

20.5 Emergency acoustic interrupt profile

A public-safety acoustic adapter may detect an emergency-signal candidate, estimate direction and confidence, and request a bounded attention interrupt. It **MUST** retain false-positive/false-negative metrics, avoid treating loudness alone as authority, and defer operational decisions to the application and human command structure.

21

Self-reference, stability, and failure exposure

21.1 Closed update

The feedback operator updates addressed cells, branch metadata, histories, quality flags, and output residuals:

$$L_{n+1} = \mathcal{U}(L_n, u_n, y_n, r_n, p_n, \chi_n). \tag{21.1}$$

For numeric state a relaxed update may be

$$L_{n+1}^{\mathrm{num}} = (1-\mu)L_n^{\mathrm{num}} + \mu\widetilde{L}_{n+1}^{\mathrm{num}}, \qquad 0 < \mu \le 1, \tag{21.2}$$

while discrete allocation and mode fields follow declared transition rules.

21.2 Fixed point and hybrid modes

For stationary input x, a self-consistent state $L^\star$ satisfies

$$L^\star = \mathfrak{N}_{\Delta t}(x; L^\star). \tag{21.3}$$

In a smooth mode, local behavior can be studied through

$$J_L = \left.\frac{\partial \mathfrak{N}_{\Delta t}(x; L)}{\partial L}\right|_{L=L^\star}. \tag{21.4}$$

A sufficient local condition is spectral radius $\rho(J_L) < 1$. The full engine is hybrid and nonsmooth because parity, mode, contact, and allocation events change the update map. Each mode and transition surface must be analyzed separately.

21.3 Boundary restoration

For one projection step with small residual,

$$F_i(\mathcal{B}_0(z)) = (1-\lambda_B)F_i(z) + O(F_i(z)^2) \tag{21.5}$$

when $\nabla F_i \neq 0$ and regularization is small. Thus $0 < \lambda_B < 2$ contracts a sufficiently small local residual to first order. This is the precise form of "pulling the apex back to the boundary."

21.4 Practical stability envelope

A run is within its declared envelope only when

$$\max_i r_{B,i} \le \tau_B, \qquad N_n \le N_{\max}, \tag{21.6}$$

$$\max_i \|v_i\| \le v_{\max}, \qquad \max_i f_{p,i} \le f_{\max}, \tag{21.7}$$

$$\|r_n\| \le \tau_r, \qquad C_n \le D_n - \Delta_{\mathrm{reserve}}, \tag{21.8}$$

$$s_n \le \tau_s, \qquad h_{\mathrm{contact}} \ge 0 \text{ or contact mode is active.} \tag{21.9}$$

No topology, coordinate transform, or parity bit guarantees this envelope.

21.5 Failure is an output

When an assumption fails, the engine produces a quality/failure state rather than silently substituting a metaphor. Mandatory failure classes include:

- invalid or singular field gradient;
- log-radius or angle saturation;
- transform pole or inverse failure;
- phase unwrap or lock failure;
- parity-rate chatter;
- branch or memory pressure;
- contact-rate bound invalidation;
- photometric saturation or missing sensor data;
- deadline miss;
- provenance-signature or clock failure; and
- application residual outside tolerance.

22

Tamper-evident timeline and identity handling

22.1 From blackboard metadata to event records

The commute transcript treats signs, tags, reflections, photographs, and timestamps as "blackboard" metadata. Version 0.3 retains that append-only event model. A canonical event record is

$$E_n = \left(g_n, T_n, d_n^{\text{input}}, d_n^{\text{state}}, d_n^{\text{output}}, m_n, r_n, \chi_n, a_n^{\text{attach}}, k_n^{\text{actor}}\right), \tag{22.1}$$

where each d is a cryptographic digest, m_n is runtime mode, χ_n contains alarms/quality, a_n^{attach} is an optional attachment digest, and k_n^{actor} is a privacy-preserving key or role identifier.

22.2 Hash chain

With a collision-resistant hash H and an unambiguous canonical encoding canon,

$$h_0 = H(\texttt{NHDF-v0.3-genesis} \parallel r_0), \tag{22.2}$$

$$h_n = H(\texttt{NHDF-event} \parallel h_{n-1} \parallel \text{canon}(E_n)). \tag{22.3}$$

The domain tags prevent cross-protocol ambiguity. A checkpoint may be digitally signed:

$$\sigma_n = \text{Sign}_{sk}(h_n \parallel g_n \parallel T_n). \tag{22.4}$$

A trusted timestamp authority or secure clock may be used when legal or evidentiary timing matters.

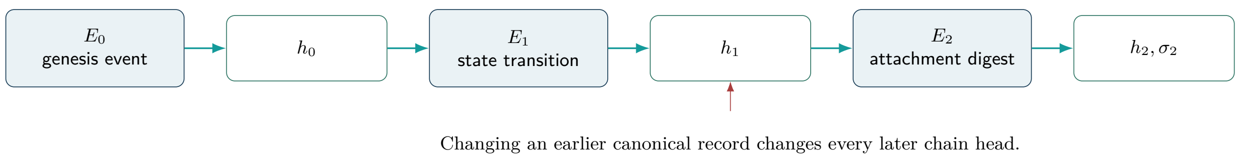

Figure 22.1: Tamper-evident forward timeline. Hash chaining detects modification when the chain head or signature is trusted.

22.3 Attachments and physical observations

A photograph, handwritten tag, location observation, or sensor snapshot becomes evidentiary only when the capture process, file bytes, timestamp, device identity, custody, and signature are controlled. The engine may store its attachment digest and context, but it may not claim that a visible sign by itself is a cryptographic proof or that a generated caption authenticates a location.

22.4 Identity data

The author declaration in this document is publication metadata. A deployed engine uses a protected private key, hardware-backed credential, or organizational certificate. It stores a pseudonymous public-key identifier where possible and separates identity from operational data.

Identity and cryptography

A name, birth date, civil identifier, vehicle number, officer number, or precinct label **MUST** not be used directly as a secret key or entropy source. A system that needs identity binding **MUST** use an authenticated credential and a documented key lifecycle, and **SHOULD** minimize exposure of raw personal identifiers.

22.5 What the provenance profile proves

A valid chain can show that a sequence of canonical records has not changed relative to a trusted chain head and signature. It does not prove that a sensor was truthful, a human interpretation was correct, a location was genuine, or an algorithm was fair. Those claims require independent controls and evidence.

23

Public-safety and responsible autonomy profile

23.1 Purpose

The author's dedication motivates a public-safety profile for police and emergency-service contexts. The profile is limited to decision support, collision avoidance, bounded sensor processing, evidence integrity, emergency-signal localization, and accountable control. It does not grant autonomous coercive authority.

23.2 Candidate applications

Table 23.1: Public-safety application mappings.

Application	NHDF/PCKF role	Required evidence
Vehicle collision avoidance	Log-polar field, frame compensation, adaptive step, CCD, safe stop.	Missed-contact rate, false alarms, stopping envelope, sensor degradation tests, conventional baseline.
Emergency acoustic localization	Log magnitude, array phase, wavefront delay, attention interrupt.	Direction error, confidence calibration, multipath/noise tests, unknown-source-frequency handling.
Dash/body-camera quality	Exposure shock, ROI, frame stabilization, quality flags.	Raw/reconstructed image metrics, dropped coverage, audit logs, no unsupported anti-blooming claim.
Evidence provenance	Event digests, signatures, generation tags, attachment hashes.	Key security, canonicalization tests, clock behavior, custody procedures, independent verification.
Bounded sensor fusion	Shared cell state and quality flags across modalities.	Calibration, uncertainty propagation, failure-mode coverage, human-readable explanation.
Operator training/simulation	Replayable trajectories, branch/contact modes, telemetry.	Scenario validity, instructor oversight, no representation as operational certification.

23.3 Minimum-intervention control: "digital jiu-jitsu"

The commute transcript uses martial-arts language to describe redirecting momentum rather than meeting it with brute force. The formal counterpart is a minimum-intervention safety filter. Given a nominal action u_{nom}, choose

$$u^\star = \arg\min_u \frac{1}{2}\|u - u_{\text{nom}}\|_R^2 \tag{23.1}$$

subject to actuator limits and safety constraints such as a control barrier condition

$$\dot{h}_j(x, u) + \alpha_j h_j(x) \geq 0, \tag{23.2}$$

where $h_j(x) \geq 0$ defines a safe set. If the problem is infeasible, the agent enters a documented safe state, ordinarily stop, retreat, or handover. This captures low-waste redirection while retaining explicit physics and authority limits.

23.4 Non-negotiable governance constraints

Public-safety constraint

A public-safety NHDF profile **MUST** satisfy all of the following:

1. A named human or legally accountable organization remains in command.
2. The engine does not independently authorize arrest, detention, use of force, pursuit escalation, or other coercive action.
3. Uncertainty, sensor quality, and failure state are visible to operators.
4. Data collection has a lawful purpose, documented retention, access control, and minimization policy.
5. Performance is tested across relevant populations, environments, lighting, acoustics, and mobility conditions; known disparities are reported.
6. Every safety-critical decision path is auditable and replayable within legal limits.
7. Cybersecurity includes signed software, key protection, rollback control, update provenance, and incident response.
8. A fail-safe mode exists for sensor conflict, deadline miss, resource exhaustion, integrity alarm, or model uncertainty.

23.5 Attention and surveillance boundary

The budgeted attention operator may prioritize safety-relevant regions, but it **MUST** not silently become unrestricted person tracking. Public-safety deployments declare the task, target class, retention period, access rights, and prohibited inferences. Where less intrusive sensing meets the safety objective, it is preferred.

23.6 No endorsement claim

The dedication and application profile do not state that Dutch police, the Hoofddorp police station, Roads Technology, ProBiblio, FLX Space, or any other named place or organization reviewed or endorsed this work.

24

Reference software and GPU realization

24.1 Semantics-first implementation order

The Version 0.3 implementation order is deliberately certification-first:

1. deterministic scalar CPU reference for query records, statuses, tolerances, and replay;
2. exact or inclusion-tested analytic primitive backends;
3. conservative swept broad phase with an auditable no-false-negative test corpus;
4. capability registry and explicit `UNSUPPORTED`/`INDETERMINATE` paths;
5. rigid, convex, triangle-feature, deformable, and implicit backends, each with isolated verification;
6. contact grouping, event-driven integration, and response as separate layers;
7. bounded parallel queues and CPU/GPU equivalence tests;
8. profiling, selective fusion, and vendor-specific optimization only after the semantic tests pass.

A fast kernel that cannot state why a pair was cleared is not a conforming general-purpose CCD backend.

24.2 Module boundaries

A reference runtime separates the following modules.

Module	**Responsibility**
Query normalizer	Validate identifiers, interval, minimum separation, tolerances, units, motion class, shape capabilities, and work budget.
Motion preprocessor	Produce conservative translational, angular, acceleration, and deformation bounds over the query interval.
Broad phase	Construct swept bounds, generate and deduplicate candidate pairs, and retain provenance for every rejection.
Capability dispatcher	Select only a backend whose declared geometry/motion domain covers the pair.
TOI backend	Return a status, a time interval, witnesses, feature identifiers, and a certificate class.
Post-checker	Re-evaluate separation on both sides of the returned interval, tighten when possible, and reject inconsistent results.
Manifold builder	Group co-time contacts, classify vertex–face/edge–edge/analytic features, and construct persistent contact identifiers.
Integrator bridge	Advance to the earliest certified event, solve or constrain response, and continue the remaining time without skipping events.
Telemetry/provenance	Record bounds, candidates, backend, interval, predicates, work, fallback, and result digest.

24.3 Structure-of-arrays state

A practical CPU/GPU runtime separates:

- object identifiers, shape type, motion type, material, collision group, and generation;
- transform endpoints, linear/angular velocity bounds, acceleration bounds, and deformation envelopes;
- primitive parameters, support-map handles, mesh vertices/edges/faces, BVH nodes, and implicit-field contracts;
- swept AABBs or stronger bounds, candidate-pair keys, and adjacency masks;
- backend-specific work queues and bounded scratch storage;
- status, TOI lower/upper endpoints, witnesses, normals, feature IDs, certificate class, and diagnostic flags;
- manifold, island, impulse/barrier, and persistent-contact records;
- NHDF payload/parity/phase/log-polar state, application output, and provenance staging.

The log-polar LUT may prioritize work, index cached states, or encode multiscale features. It does not replace the Cartesian or configuration-space bounds required by the selected CCD certificate.

24.4 Reference execution schedule

For a requested interval $[t_0, t_1]$, a semantics-first schedule is:

1. validate and normalize all query records;
2. compute conservative motion envelopes and swept bounds;
3. update/refit the broad-phase hierarchy;
4. generate, filter, deduplicate, and capacity-check candidate pairs;
5. classify initial contact or overlap at t_0;
6. dispatch remaining candidates by shape and motion capability;
7. run backend solvers and post-check returned enclosures;
8. reduce to the earliest event interval and form a co-time bucket;
9. build manifolds/contact islands and apply the declared response or barrier coupling;
10. advance the world to a safe time, update persistent contacts, and repeat for the remaining interval;
11. expose incomplete work, unsupported pairs, overflow, or numerical failure rather than treating them as clear;
12. write telemetry and the event-chain record.

24.5 Reference pseudocode

Listing 24.1: Semantics-first general-purpose CCD world step.

```
def ccd_world_step(world, t0, t1, policy):
    remaining = [t0, t1]
    events = 0

    while remaining.length > policy.time_tolerance:
        queries = normalize_queries(world, remaining, policy)
        envelopes = bound_motion_and_deformation(queries)
        swept = make_conservative_swept_bounds(queries, envelopes)

        pairs, broad_status = broad_phase_complete(swept, policy)
```

```
    if broad_status is not COMPLETE:
        return fail_world_step(BROAD_PHASE_INCOMPLETE, telemetry())

    results = []
    for pair in pairs:
        initial = classify_initial_state(pair, remaining.start, policy)
        if initial.status in {INITIAL_CONTACT, INITIAL_OVERLAP}:
            results.append(initial)
            continue

        backend = registry.dispatch(pair.capabilities)
        if backend is None:
            results.append(result(UNSUPPORTED, pair=pair))
            continue

        r = backend.time_of_impact(pair, remaining, policy)
        r = postcheck_or_downgrade(pair, r, policy)
        results.append(r)

    if any(r.status in FATAL_OR_UNRESOLVED for r in results):
        return fail_world_step(select_failure(results), telemetry(results))

    earliest = certified_earliest_event(results, policy)
    if earliest is None:
        advance_world(world, remaining.end)
        return success(CLEAR_CERTIFIED, telemetry(results))

    bucket = group_cotime_contacts(results, earliest, policy)
    advance_world(world, safe_precontact_time(earliest, policy))
    manifolds = build_contact_manifolds(bucket, world, policy)

    solve_status = couple_response_or_barrier(world, manifolds, policy)
    if solve_status is not SOLVED:
        return fail_world_step(solve_status, telemetry(results, manifolds))

    remaining.start = progress_time(earliest, policy)
    events += 1
    if events > policy.max_events_per_step:
        return fail_world_step(EVENT_ACCUMULATION, telemetry())

return success(HIT_CERTIFIED, telemetry())
```

The production implementation may batch and fuse stages, but its externally visible result must be equivalent to this fail-closed control flow for the declared profile.

24.6 Parallel and GPU mapping

The principal parallel work units are swept-bound construction, BVH refit/traversal, pair deduplication, feature-pair generation, homogeneous backend batches, interval subdivision, and contact reduction. A practical queue sequence is

$$Q_{\text{object}} \to Q_{\text{pair}} \to \{Q_{\text{primitive}}, Q_{\text{convex}}, Q_{\text{VF}}, Q_{\text{EE}}, Q_{\text{implicit}}, Q_{\text{fallback}}\} \to Q_{\text{hit}} \to Q_{\text{manifold}}.$$

Every queue has a declared capacity and overflow result. Warp-level early exits may improve throughput, but they may not convert an unresolved lane into a clear result. Parallel minimum-time reduction compares intervals and certificate priority, not only floating-point midpoints.

24.7 Memory budget

Let M_V be measured usable memory after display, driver, runtime, and fragmentation reserve. A conforming layout satisfies

$$\begin{aligned} &M_{\text{LUT}} + M_{\text{object}} + M_{\text{geometry}} + M_{\text{BVH}} + M_{\text{pair}} + M_{\text{backend}} + M_{\text{contact}} \\ &\qquad + M_{\text{timeline}} + M_{\text{output}} + M_{\text{provenance}} + M_{\text{reserve}} \le M_V. \end{aligned} \tag{24.1}$$

The source's illustrative 12 GB split remains a traceability example only. Version 0.3 requires measured peak allocation, queue high-water marks, deterministic overflow behavior, and a reserve that is not counted as application capacity.

24.8 Bitwise, matrix, and ray-query hardware

Population count and XOR naturally accelerate parity. Shifts may implement powers of two and fixed-point scaling. Matrix units may accelerate batched transforms or learned adapters. Ray-query or traversal hardware may help candidate generation when its semantics and conservativeness are documented. None of these facilities automatically implements topology, Möbius inversion, CCD certification, contact response, or proof. Low-level instructions are selected only after their exposed semantics match the reference operation.

24.9 Accompanying executable scaffold

The Version 0.3 archive contains a standard-library Python scaffold. It implements typed query/results, swept-AABB broad phase, analytic sphere–sphere and sphere–plane time of impact, moving AABB slab CCD, a bounded conservative-advancement example for an implicit sphere, explicit unsupported results, JSON test vectors, telemetry schema, and unit tests. It is a conformance seed and differential-test oracle for those elementary backends; it is not represented as the complete convex/mesh/deformable production engine specified in the preceding chapters.

25

Optical and quantum operator-equivalence extension

25.1 Physical transfer criterion

For an optical device with input field E_{in} and output E_{out}, an operator-equivalent stage satisfies

$$E_{\text{out}} = \mathcal{T}_{\text{phys}}(E_{\text{in}}) \approx \mathcal{O}(E_{\text{in}}) \tag{25.1}$$

within measured error, where $\mathcal{O}$ is a declared subset of the software chain. Phase, amplitude, polarization, spatial mode, and propagation distance may encode state, but the experiment must specify calibration, bandwidth, drift, energy, and repeatability.

25.2 Quantum criterion

A quantum state may encode an address/state superposition

$$|\Psi\rangle = \sum_i c_i |i\rangle |z_i\rangle. \tag{25.2}$$

A unitary/channel/measurement sequence must reproduce a declared operator and readout distribution within error. A classical local-zero-set constraint is not automatically a quantum error-correcting code, and the use of “holographic” does not imply quantum advantage.

25.3 Minimum physical experiment

A disciplined first experiment implements only one or two stages:

1. encode a known residual into optical or quantum coordinates;
2. apply a programmable transform or zero-set projection surrogate;
3. compare the measured output with the numerical reference;
4. report transfer error, drift, bandwidth, latency, energy, calibration, and failure rate.

Only measured operator equivalence justifies moving feedback, branching, or contact into the substrate.

26

Application adapters

26.1 Synthetic-aperture radar

The SAR profile maps a complex echo or estimated phase-error vector to the common state. Residual magnitude and phase enter the log-polar address; local zero sets represent valid focus or phase-consistency states; the router partitions range, azimuth, subaperture, scale, or hypothesis space; frame state represents platform motion; the wavefront stage models antenna support; and projection produces a focused image or correction field.

A severe ionospheric or hardware disturbance may raise jitter, phase acceleration, quantizer saturation, parity rate, and residual. The formal response is bounded safe mode, stronger integrity checking, uncertainty reporting, and comparison with a conventional autofocus baseline. The profile makes no claim of geomagnetic-storm immunity.

Minimum SAR metrics include phase-error RMS, image entropy or sharpness, point-target resolution, peak and integrated sidelobe ratios, runtime, memory, power, dropped frames, saturation, parity behavior, and failure under even-numbered bit faults.

26.2 Rendering and spatial simulation

A rendering adapter may use local fields, analytic cone sweeps, log-polar foveation, and the CCD profile. It reports image error, missed intersections, contact error, temporal stability, memory, and frame-time distribution. “Infinite detail” is approximated by bounded recursion, finite precision, and finite samples.

26.3 Robotics and navigation

A robotics adapter uses observer-frame compensation, conservative contact, minimum-intervention control, and safe mode. It separates perception uncertainty from dynamics and preserves an externally verifiable stop/retreat behavior. Soft-robot or articulated-tree mappings are candidate models, not structural identities.

26.4 Audio and spatialization

An audio adapter may route frequency-band states through the branch graph and map wavefront direction to output channels. Jitter and phase may modulate timbre, but signal quality, stability, clipping, latency, and hearing-safety constraints remain explicit.

26.5 Edge AI adapter

An AI adapter may use log-polar indexing, sparse cell activation, branch routing, or implicit parameterization. It does not follow that dense model weights disappear or that a model too large for memory will run automatically. Compression ratio, perplexity/accuracy, latency, energy, memory, calibration, and baseline quantization must be measured.

27

Verification program and falsifiable hypotheses

27.1 CCD conformance ladder

General-purpose claims are separated into evidence levels so that an elementary implementation cannot inherit the claims of the complete architecture.

Table 27.1: Version 0.3 CCD conformance ladder.

Level	Name	Minimum evidence
G0	Contract model	Query/result types, status lattice, units, tolerances, capability registry, and deterministic replay compile and pass schema tests.
G1	Elementary analytic	Initial-state handling and certified sphere–sphere, sphere–plane, point/segment, or moving-AABB cases pass analytic and high-precision references.
G2	Rigid broad/convex	Conservative swept broad phase plus a validated convex or rigid conservative-advancement backend; rotation and angular bounds included.
G3	Surface mesh	Complete vertex–face and edge–edge candidate generation, certified/inclusion TOI, degeneracy handling, self-collision filtering, and manifold construction.
G4	Deformable/implicit	Declared deformable surface/volume and implicit-field backends with motion/field error bounds, topology-change policy, and fail-closed unsupported paths.
G5	Integrated world step	Earliest-event reduction, simultaneous contacts, response/barrier coupling, joints, persistent contacts, bounded event accumulation, CPU/GPU equivalence, and release-level stress tests.

An implementation reports the highest level it actually demonstrates and lists which backend domains remain unsupported.

27.2 Core falsifiable hypotheses

Table 27.2: Version 0.3 falsifiable hypotheses.

ID	Hypothesis	Minimum test
H1	Log-polar addressing reduces weighted quantization error for a declared residual distribution relative to an equal-capacity linear or Cartesian table.	Matched-capacity Monte Carlo with saturation, inverse-map error, and confidence intervals.
H2	Jacobian-correct log-polar kinematics has lower physical trajectory error than independent radial/angular shifts.	Pure radial, pure angular, coupled, near-apex, and high-angular-speed trajectories.
H3	Local zero-set projection contracts residual without destabilizing the integrator.	Sweep relaxation, field conditioning, time step, and noise; report singular/fallback cases.

ID	Hypothesis	Minimum test
H4	A conservative swept broad phase generates every pair whose exact swept shapes approach within $d_{\min}$.	Exhaustive small scenes, randomized high-precision oracle, adversarial thin/fast/rotating cases, and pair-rejection audit.
H5	Every `CLEAR_CERTIFIED` result is a true no-contact result over the complete requested interval and separation margin.	Independent high-precision sampling plus analytic/inclusion oracle; zero false clears in the release corpus.
H6	Every certified hit interval contains the first admissible contact and contracts monotonically with additional work.	Compare returned enclosures at increasing budgets against analytic or interval ground truth.
H7	Rigid conservative advancement remains conservative when translation, rotation, and acceleration obey the declared bounds.	Long thin rotating bodies, off-center features, curved motion, changing closest features, and tight angular-radius bounds.
H8	Vertex–face and edge–edge mesh backends do not miss first contact under linear vertex trajectories.	Coplanar, near-parallel, tangential, endpoint, zero-area, high-aspect, and high-dynamic-range cases.
H9	Deformable/self-collision filtering preserves all non-adjacent physical candidates while removing only declared topological exclusions.	Cloth, tetrahedral, articulated, and self-folding sequences with adjacency mutations.
H10	Initial overlap, initial contact, grazing, and separated motion are classified distinctly and reproducibly.	Exact boundary starts, shallow penetration, zero relative normal speed, and tolerance sweeps.
H11	Co-time grouping and manifold construction reduce event-order dependence without merging temporally distinct impacts.	Symmetric multi-body impacts, corner/edge strikes, stacked contacts, and permutation tests.
H12	Event-driven integration makes positive progress or returns a bounded accumulation failure; it does not spin indefinitely.	Resting contacts, bouncing stacks, chatter, Zeno-like sequences, and very small remaining intervals.
H13	The response layer matches its declared restitution/friction or barrier model without being used to hide detection errors.	Momentum/energy residual, friction cone, no-penetration, joint consistency, and response-disabled replay.
H14	CPU and GPU implementations return equivalent status/certificate classes and compatible intervals within declared tolerance.	Deterministic replay across queue order, block size, precision mode, overflow, and backend fallback.
H15	The status and failure lattice prevents unsupported, overflowed, or numerically unresolved work from appearing as clear.	Fault injection at every queue, iteration, predicate, and field-contract boundary.
H16	Budgeted attention and NHDF routing improve work allocation without violating the broad-phase completeness floor.	Equal-safety comparison with uniform scheduling; verify that prioritization changes latency, not candidate completeness.
H17	Hash-chained telemetry detects record mutation/reordering relative to a trusted chain head.	Canonicalization, mutation, truncation, rollback, and key-rotation tests.
H18	Any claimed application advantage survives ablation of the NHDF-specific operators at equal correctness and resource limits.	Preregistered application benchmark against simpler CCD/runtime baselines.

27.3 Ground truth and acceptance semantics

A floating-point timestamp is not sufficient ground truth. For a query with true first-contact time t^*, the preferred oracle is an analytic value or an independently generated enclosure $I^* = [t^*_L, t^*_U]$. A returned certified hit interval $I = [\underline{t}, \bar{t}]$ passes when

$$I \cap I^* \neq \varnothing, \qquad \underline{t} \leq t^* \leq \bar{t} \tag{27.1}$$

when t^* is known, and its pre-contact witness satisfies the declared positive-separation test. A certified clear passes only when the oracle proves no admissible contact over the full interval. An approximate hit may be conservative, but an approximate clear is never promoted to certified clear.

The release corpus uses four verdict classes:

Sound pass The result and certificate meet the relevant mathematical obligation.

Conservative pass Contact is reported no later than allowed and no false clear occurs, but the interval or witness is looser than the target.

Declared unresolved The backend returns `INDETERMINATE`, `UNSUPPORTED`, overflow, or numeric failure exactly where its contract permits.

Unsound failure A false clear, excluded true TOI, invalid witness/normal presented as certified, silent queue loss, or non-terminating world step.

27.4 Mandatory adversarial test families

27.4.1 Elementary analytic and scale tests

The corpus includes head-on, receding, tangent, double-root, initial-contact, and initial-overlap sphere pairs; moving sphere/plane; point/segment and capsule-like reductions; moving AABBs; zero velocity; very small and very large scales; and unit-rescaling invariance. Analytic roots are computed with cancellation-resistant quadratic formulas and independently checked in extended precision.

27.4.2 Broad-phase completeness

Mandatory scenes include a fast small object crossing a thin static object; two rotating long rods whose endpoint swept bounds overlap; diagonal motion through a grid; identical bounds; near-touch at $d_{\min}$; objects entering from outside the current spatial hash; hierarchy refit after large motion; duplicate-pair stress; and queue-at-capacity. Every exact/narrow-phase hit must be represented in the candidate log.

27.4.3 Rigid convex motion

Tests span pure translation, pure rotation, screw motion, high angular speed, off-center contact, changing closest features, nearly parallel faces, flat/slender shapes, support-map discontinuity, nonuniform scale rejection, and acceleration/trajectory envelopes. The suite compares analytic or high-precision shape casting where available and requires conservative fallback otherwise.

27.4.4 Triangle and polygonal surface CCD

The elementary mesh suite enumerates vertex–face and edge–edge first contacts, endpoint/endpoint reductions, coplanar motion, grazing, nearly parallel edges, zero-length edges, zero-area triangles, shared-feature exclusions, high-aspect triangles, mirrored winding, duplicate vertices, and large-coordinate/small-gap combinations. The solver must expose which degeneracy path or predicate decided the result.

27.4.5 Deformable, self-collision, and topology change

Required sequences include folding cloth, an edge passing a vertex near an adjacent face, tetrahedral compression, articulated chains, rapidly changing deformation bounds, disabled adjacent pairs, newly created topology, deleted features, and remeshing during a step. A topology change either has a declared continuous map and rebuilt structures or becomes a discrete boundary at which the interval is split.

27.4.6 Implicit and sampled fields

Tests cover exact sphere/box fields, conservative Lipschitz fields, scaled non-distance implicit functions, voxel/trilinear fields with known interpolation error, moving fields, discontinuous edits, gradient singularities, thin features below cell size, and deliberately invalid contracts. A field lacking a conservative distance/error/rate certificate must not return `CLEAR_CERTIFIED` through the implicit backend.

27.4.7 Simultaneous and persistent contact

The corpus includes a sphere striking a corner, a box landing flat, symmetric three-body impact, stacked rigid bodies, a sliding block, rolling contact, joints plus contact, and several contacts whose TOIs differ just inside/outside the co-time tolerance. Reordering object IDs, pair queues, or thread blocks must not alter the declared deterministic profile beyond the permitted tolerance.

27.4.8 Resource and failure injection

Each finite resource is tested at zero, one below capacity, exactly capacity, and one above capacity. Injected faults include stale generation, NaN/Inf, predicate uncertainty, iteration exhaustion, invalid motion bound, unsupported shape pair, malformed mesh, broad-phase overflow, backend-queue overflow, event-limit exhaustion, and post-check inconsistency. The expected result is explicit failure or a declared conservative fallback, never silent success.

27.5 Differential, metamorphic, and fuzz testing

The verification harness should combine:

- differential comparison between analytic, inclusion, conservative-advancement, and brute-force/high-precision oracles on overlapping domains;
- randomized scene generation biased toward small gaps, double roots, near-parallel features, high dynamic range, and changing closest features;
- metamorphic transforms: translation, rigid rotation, object-label permutation, time scaling, uniform unit scaling, mirrored coordinates, and interval subdivision;
- shrinking of every failing random case to a minimal reproducible query record;
- replay under multiple precision modes, compiler optimization levels, CPU architectures, and GPU work schedules;
- corpus retention with stable IDs and expected status/certificate classes.

A metamorphic transform is accepted only when its mathematical preconditions preserve the query; nonuniform scaling, for example, changes several backends and must not be treated as a free invariance.

27.6 Performance and scaling measurements

Correctness is primary, but a general-purpose architecture must publish cost. Report:

- objects, primitives/features, candidate pairs, accepted backend pairs, and contact count;
- broad-phase build/refit/traversal time and candidate amplification ratio;
- narrow-phase work distribution, interval subdivisions, predicates, and fallback rate;
- CPU/GPU latency distribution, throughput, memory peak, queue high-water marks, and energy when available;
- scaling with object count, mesh resolution, deformation speed, angular speed, contact density, and requested tolerance;

- worst-case and tail latency, not only average frame time.

Performance comparisons use the same collision margin, failure policy, and soundness target. A faster algorithm that silently misses contacts is not a valid baseline win.

27.7 Required ablations

At minimum, experiments remove or replace one component at a time:

- log-polar metric correction versus raw shifts;
- local zero-set projection versus unconstrained state;
- NHDF attention/routing versus neutral scheduling at identical candidate completeness;
- swept broad phase versus discrete endpoint broad phase;
- capability dispatch versus one universal approximate backend;
- analytic primitive backend versus conservative advancement;
- interval/inclusion root isolation versus sampled or Newton-only localization;
- exact/filtered predicates versus floating-point-only classification;
- angular/acceleration bounds versus translation-only advancement;
- vertex–face and edge–edge decomposition versus discrete mesh checks;
- co-time manifold grouping versus strictly serial contact order;
- event-driven response versus advance-then-project overlap repair;
- open-loop output versus self-referential NHDF feedback.

If a simpler ablation performs equally well at the same correctness and resource constraints, the removed operator is not justified for that profile.

27.8 Telemetry contract

Every conforming CCD query records at least

$$\begin{aligned}\{&\text{query ID}, t_0, t_1, d_{\min}, \tau_t, \tau_x, \text{shape classes}, \text{motion classes},\\ &\text{swept-bound type}, \text{candidate/rejection reason}, \text{backend}, \text{status}, \gamma, [\underline{t}, \overline{t}], \text{feature IDs},\\ &\text{witnesses}, \text{normal validity}, \text{iterations/subdivisions/predicates}, \text{fallback}, \text{queue occupancy}, \text{overflow},\\ &\text{post-check residual}, \text{manifold/island ID}, \text{response status}, \text{runtime}, \text{generation}, h_n\}. \end{aligned} \tag{27.2}$$

The wider NHDF run additionally records log-polar address, boundary residual, parity/phase state, transform flags, attention budget, application residual, saturation, and mode transitions.

27.9 Release gates

A release claiming a Version 0.3 general-purpose CCD level must satisfy all of the following for the declared domain:

1. zero known false `CLEAR_CERTIFIED` results in the mandatory, regression, and randomized release corpus;
2. every certified hit interval passes the independent enclosure/post-check criteria;
3. every unsupported or unresolved query is exposed through the status lattice;
4. broad-phase candidate completeness is audited against the oracle corpus;

5. all finite-resource overflow paths terminate and report failure deterministically;
6. world steps terminate with success or a bounded failure; no hidden retry loop remains;
7. CPU/GPU status and certificates agree under the published equivalence policy;
8. documentation publishes backend domains, tolerance policy, known exclusions, corpus version, and test digest.

These gates are evidence requirements, not a claim that testing proves correctness for every possible real-valued input.

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Limitations and failure modes

Honest scope of Version 0.3

Version 0.3 specifies how a general-purpose engine should cover many common rigid, convex, mesh, deformable, compound, and implicit cases without silently misclassifying unknown work. It does not prove that one implementation already handles every geometry, motion, numerical scale, or discontinuity. The normative improvement is explicit capability, certification, and failure semantics.

28.1 Fundamental CCD limits

1. **No single universal closed-form solver.** General-purpose CCD is a dispatch architecture over geometry, motion, and certificate capabilities. Arbitrary procedural shapes or motions without computable bounds may remain unsupported.
2. **Continuity is an assumption.** Teleports, discontinuous deformation, instantaneous topology edits, or unbounded control inputs cannot be covered by a continuous trajectory proof unless the interval is split at the discontinuity.
3. **Initial overlap is not time of impact.** Penetrating input has no positive pre-impact interval. It requires overlap classification and a separate recovery or barrier policy.
4. **Detection is not response.** A correct TOI does not determine restitution, friction, compliance, fracture, or stable stacking. Conversely, projection or impulse response cannot repair a missed first contact certificate.
5. **Simultaneous contact is tolerance-dependent.** Exact simultaneity is unstable under finite precision. Co-time grouping can reduce order dependence but may merge or separate events near its threshold.
6. **Event accumulation.** Resting contact, chatter, and Zeno-like sequences can produce many events in a finite interval. The engine needs persistence, resting models, progress floors, or an explicit accumulation failure.

28.2 Geometry and motion limits

7. **Broad-phase completeness depends on conservative bounds.** A too-small swept AABB, stale BVH, incorrect rotational envelope, or dropped queue entry can create false negatives before the narrow phase runs.
8. **Rotation and acceleration bounds may be loose.** Support-radius angular bounds are conservative but can reduce performance severely for long thin shapes or large rotations.
9. **Concave and compound objects require decomposition.** A backend that handles convex pieces does not automatically handle arbitrary concave solids unless piece coverage and pair aggregation are complete.
10. **Triangle feature decomposition has degeneracies.** Coplanar, parallel, zero-area, zero-length, duplicated, and shared-feature cases require explicit predicates and reduction rules.
11. **Deformable validity depends on motion envelopes.** Linear vertex trajectories are not a proof for higher-order or solver-coupled deformation unless the approximation error is bounded.

12. **Topology changes invalidate cached features.** Remeshing, fracture, activation, or deletion can invalidate adjacency, persistent IDs, BVHs, and contact manifolds.
13. **Implicit fields need a contract.** A generic implicit residual is not a signed distance. Conservative advancement requires a valid lower separation bound, Lipschitz/gradient bound, interpolation error, and time-rate bound.
14. **Thin sampled features can disappear.** Voxel or neural fields may not represent geometry below sampling or approximation scale; no traversal logic can recover missing geometry without an external conservative envelope.
15. **Non-orientable/global-sign ambiguity.** Intrinsic charts or unsigned/local constraints are required where a global inside/outside sign does not exist.

28.3 Numerical and certification limits

16. **Floating-point roots are not automatically certified.** Newton or polynomial formulas can lose roots through cancellation, tangency, or classification error. Filters, exact predicates, interval arithmetic, and post-checks reduce but do not erase implementation risk.
17. **Tolerance changes the problem.** Time, distance, feature, and co-time tolerances define an expanded/-contracted collision condition. They must scale with units and must not be hidden as "epsilon."
18. **Exact predicates do not make all quantities exact.** Topological signs may be exact while witnesses, normals, and times remain interval or floating-point approximations.
19. **Conservative does not mean tight.** A conservative hit may occur early. Excessively loose bounds can freeze motion or generate too many candidate contacts.
20. **Post-checks can discover inconsistency but may not diagnose its source.** Motion bounds, geometry, predicates, or queue corruption may each cause failure.
21. **Determinism is a profile.** Parallel reductions, floating-point associativity, atomics, and dynamic queues can produce different but still conservative intervals unless ordering is controlled.

28.4 NHDF and systems limits

22. **Degenerate field risk.** A literal global $F \equiv 0$ supplies no usable normal or distance.
23. **Log-polar singularity.** The apex requires a finite clamp, sentinel, or alternate chart.
24. **Metric omission.** Raw address shifts create radius-dependent physical error and omit centripetal/cross terms.
25. **Transform poles and encoding mismatch.** Möbius or hyperbolic operators can diverge; bit instructions may not implement their mathematics.
26. **Parity blind spots.** One-bit parity misses even fault counts and corrects nothing.
27. **Phase noise and hybrid chattering.** $\Delta^2\phi$ amplifies noise; parity, phase, attention, shock, and contact modes require hysteresis and rate limits.
28. **Finite-resource incompleteness.** Pair, backend, manifold, and event queues require hard capacity. Overflow makes the step unresolved unless a complete fallback is available.
29. **Hardware mismatch.** Matrix, ray, optical, or quantum units help only when exposed operations match the certified reference semantics.
30. **Attention cannot prune safety candidates.** NHDF foveation may prioritize work, but a general-purpose safety profile cannot drop a collision pair solely because it is peripheral or low-saliency.
31. **Sensor physics.** Coordinate transforms cannot reconstruct clipped samples, prevent physical sensor blooming, or remove all rolling-shutter artifacts.

32. **Provenance scope.** A hash chain proves consistency relative to a trusted head, not truth of geometry, motion bounds, or the captured world.
33. **Identity/privacy risk.** Raw personal data is neither cryptographic entropy nor correctness evidence and can create legal/security liabilities.
34. **Topology is not resilience.** A Klein-bottle quotient does not protect hardware from radiation, malware, memory faults, or corrupted inputs.
35. **Application and institutional dependence.** Rendering, robotics, SAR, audio, and public-safety profiles require independent domain validation, legal authority, cybersecurity, human factors, and oversight beyond this specification.

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Normative minimum specification, Version 0.3

An implementation may call itself an NHDF/PCKF Version 0.3 engine only when it satisfies the core requirements below. It may claim the *general-purpose CCD profile* only when it also satisfies the CCD requirements for its published conformance level and domain.

29.1 Core NHDF requirements

K1. It **MUST** define a monotonic logical timeline, generation tags, elapsed-time semantics, and ring-buffer rules.

K2. It **MUST** define causal input residuals, log-polar encoding, quantization, saturation, inverse interpretation, units, and quality flags.

K3. It **MUST** distinguish jitter log magnitude ρ_J from spatial log radius w, golden-ratio scalar φ_g from signal phase ϕ, and polar angle from phase unless a profile explicitly couples them.

K4. It **MUST** implement or explicitly bypass the log-polar Jacobian and state how address-space rates become physical motion.

K5. It **MUST** regularize the apex singularity with a finite clamp, sentinel, or alternate chart.

K6. It **MUST** represent “SDF zero” as non-degenerate local constraints or document an equivalent formulation with usable normals/separation data.

K7. It **MUST** define parity inputs, topology-orientation semantics, blind spots, debounce, and safe-mode behavior.

K8. It **MUST** define phase unwrapping/filtering, $\Delta^2\phi$, and phase-lock behavior when those operators are used.

K9. It **MUST** bifurcate only at declared branch nodes, preserve declared ordering, and bound all node allocation.

K10. It **MUST** declare exactly how φ_g and $\Delta^2\phi$ affect branch geometry, priority, or workload.

K11. Every optional coordinate transform **MUST** declare equations, units, domain, singularities, orientation effect, inverse/fallback, and a reference-equivalence test.

K12. It **MUST** integrate forward kinematics with a declared numerical method, trajectory model, and error/time-step policy.

K13. It **MUST** define the sweep/wavefront, projection, application residual, and feedback written into the next generation.

K14. It **MUST** bound cells, branches, histories, output workspace, work queues, and runtime reserve.

K15. It **MUST** emit telemetry and a visible failure state when an assumption, certificate, deadline, or resource bound is violated.

29.2 General-purpose CCD query and capability requirements

C1. Every CCD request **MUST** carry object IDs, shape/motion capability records, interval $[t_0, t_1]$, minimum separation $d_{\min}$, time/spatial tolerances, work limits, units, and policy identifier.

C2. The shape record **MUST** distinguish analytic primitive, support-mapped convex, rigid/deformable mesh, volume mesh, compound/concave decomposition, and implicit/sampled field capabilities.

C3. The motion record **MUST** identify translation, rigid rotation, articulated/deformable trajectory, interpolation order, discontinuities, and conservative speed/angular/acceleration/deformation bounds.

C4. A backend **MUST** be selected only when its published shape, motion, scale, and tolerance domain covers the query. Otherwise the result is `UNSUPPORTED` or a declared conservative fallback.

C5. The engine **MUST** classify `INITIAL_CONTACT` and `INITIAL_OVERLAP` separately from a positive-time hit and clear result.

C6. The engine **MUST** use a published status lattice including at least certified clear, certified/conservative hit, initial contact/overlap, indeterminate, unsupported, numeric failure, and resource failure.

C7. Every nontrivial result **MUST** include a certificate class and diagnostic provenance sufficient to identify the backend and assumptions used.

C8. `CLEAR_CERTIFIED` **MUST** mean no admissible contact within $d_{\min}$ over the complete requested interval under all declared motion/geometry uncertainties. Approximate or sampled evidence alone may not produce this status.

C9. A hit result **MUST** return a TOI enclosure $[\underline{t}, \overline{t}]$ or an explicit conservative point-plus-error representation, not an unexplained scalar.

C10. A backend that exhausts work, loses a queue entry, violates a bound, encounters an invalid field, or fails its post-check **MUST** not return clear.

29.3 Broad-phase and dispatch requirements

B1. Swept bounds **MUST** conservatively enclose each object's shape over the interval, including requested separation margin, rotation, acceleration, and declared deformation uncertainty.

B2. Broad-phase pruning **MUST** preserve candidate completeness: every pair capable of reaching the margin must survive or be handled by a formally equivalent aggregate test.

B3. Hierarchy refit/rebuild, spatial-hash insertion, sweep-and-prune ordering, sleeping, collision masks, and self-adjacency filters **MUST** have explicit invalidation rules.

B4. Candidate pairs **MUST** have stable IDs, be deduplicated, and be stored in bounded queues with an explicit overflow status or complete fallback.

B5. Every rejected pair **SHOULD** expose a rejection reason and bound type in audit telemetry.

B6. The dispatcher **MUST** route only to registered compatible backends. Its registry covers elementary primitives, convex pairs, triangle feature pairs, deformable or volume pairs, implicit fields, and compounds.

B7. Concave or compound coverage **MUST** state how components cover the original shape and how earliest component events are reduced.

29.4 Narrow-phase and certification requirements

N1. Analytic primitive solvers **MUST** use numerically stable root/classification logic and distinguish tangency, receding motion, initial state, and roots outside the interval.

N2. Convex/rigid conservative advancement **MUST** use a valid separation lower bound and closing-speed bound including angular and acceleration terms for the declared motion.

N3. Triangle-surface CCD **MUST** generate complete vertex–face and edge–edge candidate tests, including endpoint/degenerate reductions and declared shared-feature exclusions.

N4. Deformable backends **MUST** bound interpolation or integration error; linear-trajectory solvers may not silently cover higher-order motion without an envelope.

N5. An implicit-field backend **MUST** publish whether its value is exact distance, lower bound, or generic residual and provide Lipschitz/gradient, interpolation, and time-rate/error bounds sufficient for its certificate.

N6. Root localization **MUST** use analytic, exact-predicate, interval/inclusion, controlled conservative advancement, or another published method whose no-missed-root obligation is testable.

N7. Predicate uncertainty **MUST** trigger refinement, exact fallback, conservative classification, or an unresolved status; it may not be resolved by arbitrary sign choice.

N8. Tolerances **MUST** be dimensioned, scale-aware, recorded in the query, and applied consistently to broad phase, TOI, feature classification, co-time grouping, and post-check.

N9. Every certified result **MUST** pass an independent post-check of motion, separation, interval ordering, and witnesses. Inconsistency downgrades or fails the result.

N10. Increasing the work budget **SHOULD** preserve and tighten a certified enclosure rather than excluding a previously possible true first contact.

29.5 Manifold, integration, and response requirements

R1. The engine **MUST** reduce all candidate results to the earliest admissible event using interval-aware ordering.

R2. Contacts whose intervals overlap the declared co-time window **MUST** be grouped before response when the profile claims simultaneous-contact handling.

R3. Contact records **MUST** include stable object/feature IDs, witnesses, normal validity, separation/-margin, TOI interval, and persistence key.

R4. Self-collision filtering **MUST** remove only declared adjacency/topological pairs and **MUST** rebuild or invalidate after topology change.

R5. A topology edit or motion discontinuity **MUST** split the time interval or provide a conservative continuous map; stale features may not remain certified.

R6. The integrator **MUST** advance to a safe pre-contact/contact time, apply response or barrier coupling, and process the remaining interval without skipping earlier unresolved events.

R7. The world step **MUST** guarantee positive logical progress, transition to a resting/persistent model, or return a bounded event-accumulation failure.

R8. Collision response **MUST** be a separate declared contract: impulse, complementarity, penalty/compliance, barrier, projection, or application-specific rule.

R9. Restitution, friction, joints, moving boundaries, sleeping, and overlap recovery **MUST** expose their assumptions and failure states.

R10. Fragmentation or remeshing **MUST** create bounded new geometry, rebuild collision structures, and begin a new certified interval; it may not be treated as a free XOR split.

29.6 Parallel, determinism, and evidence requirements

P1. Every object, pair, backend, subdivision, hit, manifold, and event queue **MUST** have capacity and overflow semantics.

P2. Parallel early exit, compaction, and reduction **MUST** preserve unresolved lanes and interval/certificate ordering.

P3. A deterministic profile **MUST** define object/pair ordering, tie breaks, arithmetic mode, and reduction order; a non-deterministic profile must still preserve sound status/certificates.

P4. CPU/GPU implementations **SHOULD** pass the same replay corpus and publish their interval/status equivalence policy.

P5. The implementation **MUST** publish its conformance level, backend matrix, exclusions, tolerance policy, mandatory test-corpus digest, and known failures.

P6. Release evidence **MUST** include broad-phase completeness tests, ground-truth TOI tests, degeneracy/fuzz tests, resource-failure tests, and world-step progress tests.

P7. A provenance profile **MUST** use canonical records, a cryptographic hash, protected signing keys when signatures are claimed, and privacy-preserving identity references; raw identity is not key material.

P8. A public-safety profile **MUST** comply with the human-authority, no-autonomous-force, privacy, audit, fairness, cybersecurity, uncertainty-display, and fail-safe requirements in chapter 23.

P9. Speculative optical, quantum, SAR, AI, robustness, or institutional claims **MUST** remain separate from CCD/core conformance until independently validated.

Compact formal definition

A Version 0.3 NHDF engine with the general-purpose CCD profile is a bounded causal hybrid transition

$$\mathfrak{N}_{[t_0,t_1]} : X \times L \times H \to L \times H \times \mathcal{R}$$

with predicted trajectories $\widehat{\mathcal{T}}_n$ and collision subsystem

$$\begin{aligned}
\widehat{\mathcal{T}}_n &= \mathcal{K}^J_{T\to} \circ \mathcal{R}_{\mathrm{BST}} \circ \mathcal{C}_\phi \circ \mathcal{P} \circ \mathcal{B}_0 \circ \mathcal{X}_{\xi_n} \circ \mathcal{A}_B \circ \mathcal{E}_{\mathrm{LP}} \circ \mathcal{C}_{\mathrm{in}}(x_n; L_n),\\
\mathcal{R}_n &= \mathcal{R}_{\mathrm{C}} \circ \mathcal{M}_{\mathrm{C}} \circ \mathcal{T}_{\mathrm{TOI}} \circ \mathcal{D}_{\mathrm{CCD}} \circ \mathcal{B}_{\mathrm{CCD}}(\widehat{\mathcal{T}}_n, \mathcal{G}_n, \mathcal{P}_n),\\
y_n &= \Pi \circ \mathcal{S}(\mathcal{R}_n.\text{safe trajectories}),\\
L_{n+1} &= \mathcal{U}(L_n, y_n, r_n, \mathcal{R}_n),\\
h_{n+1} &= \mathcal{H}(h_n, E_{n+1}).
\end{aligned}$$

The defining invariant is fail-closed non-crossing: no pair is reported `CLEAR_CERTIFIED` unless the complete broad/narrow pipeline certifies the requested separation over the full interval; all other uncertainty is carried as contact, conservative hit, unsupported, indeterminate, or explicit failure.

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Conclusion

Version 0.1 made the original prompt chain coherent by treating it as a typed, bounded operator specification. Version 0.2 added the useful scale metric, transform, attention, sensor, provenance, and first contact-guard material developed in the commute transcript. Version 0.3 focuses on the user's literal computer-graphics meaning of Continuous Collision Detection and replaces the single SDF/gap guard with a general-purpose contact architecture.

The essential change is architectural rather than rhetorical. A graphics engine cannot obtain general-purpose CCD from a parity flip, an endpoint sign change, or one SDF lookup. It needs conservative swept candidate generation; shape and motion capability records; backend dispatch; analytic, convex, mesh-feature, deformable, and implicit algorithms; root enclosures and robust predicates; explicit initial overlap; contact manifolds and simultaneous events; event-driven integration; declared response; bounded queues; and statuses that preserve uncertainty instead of calling it clear.

NHDF remains useful as the surrounding representation and scheduling paradigm. Its local zero sets can provide valid implicit constraints; its log-polar metric can organize multiscale state; its causal timeline and bounded branch router can schedule work; and its provenance chain can make decisions replayable. None of those operators substitutes for the certificate obligations of the CCD subsystem. In Version 0.3 they are composed with, rather than confused with, collision detection.

The resulting specification is intentionally fail-closed. It spans common primitives, support-mapped convex bodies, rigid and linearly deforming triangle meshes, compounds, and bounded implicit fields through a backend matrix. It also permits `UNSUPPORTED`, `INDETERMINATE`, and resource/numerical failure. That is not a weakness of the formalization: it is what prevents an unhandled case from becoming a false no-collision answer.

The accompanying scaffold demonstrates the elementary analytic and conservative control flow and provides machine-readable statuses, vectors, schema, and tests. The next engineering milestone is not to claim universality, but to implement and validate successive conformance levels: complete broad phase, rigid convex motion, robust mesh feature CCD, deformable/implicit profiles, and finally integrated simultaneous-contact world stepping on CPU and GPU.

Version 0.3 therefore constitutes a substantially stronger attempt at a literal general-purpose CCD solution: a finite, causal, backend-dispatched, certificate-carrying subsystem whose claims can be replayed, ablated, stress-tested, and rejected when its assumptions do not hold.

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Research references

The following primary research sources inform the Version 0.3 CCD extension. They are not part of the supplied prompt transcripts; they are external technical context used to fill the general-purpose CCD gaps.

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A

Expanded symbol table

Symbol	Meaning
$x_n, \bar{x}_n, e_n, J_n$	Input, causal baseline, residual, and residual magnitude.
ρ_J, θ, T	Log jitter magnitude, selected polar direction, and forward-time address.
w, r, r_0, b	Log spatial radius, physical radius, reference radius, and log base.
$J_{\mathrm{LP}}, G_{\mathrm{LP}}$	Log-polar Jacobian and induced metric.
L_n, c_i, q_i	Complete state, cell record, and packed payload.
a_i, v_i, α_i	Distributed apex position, velocity, and acceleration.
φ_g	Golden-ratio scalar.
$\phi_i, \Delta^2\phi_i$	Signal/cell phase and second finite difference.
$F_i, M_i, r_{B,i}$	Local implicit constraint, its zero set, and boundary residual.
$p_i^{\mathrm{data}}, p_i^{\mathrm{topo}}, \bar{p}_i$	Payload parity, orientation parity, and debounced gate.
$o_i, \kappa_i, N_{\max}$	Local orientation, BST ordering key, and branch capacity.
Q_C	Typed CCD query record.
R_C	CCD result record containing status, TOI interval, witnesses, normal, features, certificate, work, and flags.
$[t_0, t_1]$	Requested continuous-motion interval.
$[\underline{t}, \overline{t}]$	Returned enclosure of first admissible contact.
$d_{\min}$	Requested minimum separation/collision margin.
τ_t, τ_x, τ_c	Time, spatial/predicate, and co-time/contact tolerances.
γ	Certificate class: exact predicate, inclusion, controlled conservative advancement, analytic bounded, or approximate.
B_F	Conservative upper bound on closing rate of a separation lower bound.
R_A, R_B	Support radii used in angular closing-speed bounds.
$\mathcal{G}_n$	Geometry, motion, hierarchy, and collision-policy state for generation n.
$\mathcal{P}_n$	CCD tolerance, work-budget, determinism, and response policy.
$Q_{\mathrm{pair}}, Q_{\mathrm{VF}}, Q_{\mathrm{EE}}$	Candidate-pair and triangle vertex–face/edge–edge work queues.
Γ_C	Contact manifold or co-time contact set.
$\mathcal{C}_{\mathrm{in}}$	Calibration, photometric, and observer-frame conditioning.
$\mathcal{E}_{\mathrm{LP}}$	Log-polar residual/address operator.
$\mathcal{A}_B$	Budgeted attention/resource allocator.
$\mathcal{M}_{c,R}, \Gamma_\kappa, \mathcal{G}_c$	Möbius inversion, log-spiral shear, and optional gyrotranslation profiles.
$\mathcal{B}_0, \mathcal{P}, \mathcal{C}_\phi$	Local zero-set projection, parity reduction, and phase/control operator.
$\mathcal{R}_{\mathrm{BST}}, \mathcal{K}_{T\to}^J$	Bounded branch router and forward physical kinematics.
$\mathcal{B}_{\mathrm{CCD}}$	Conservative swept broad phase.
$\mathcal{D}_{\mathrm{CCD}}$	Capability-based narrow-phase backend dispatcher.
$\mathcal{T}_{\mathrm{TOI}}$	Time-of-impact localization and certificate operator.
$\mathcal{M}_{\mathrm{C}}$	Co-time grouping, feature classification, manifold, and contact-island operator.
$\mathcal{R}_{\mathrm{C}}$	Declared impulse/barrier/contact-coupling operator.
$\mathcal{C}_{\mathrm{GP}}$	Complete general-purpose CCD composition.
$\mathcal{S}, \Pi, \mathcal{U}$	Wavefront/implicit sweep, observation projection, and self-referential update.

Symbol	Meaning
$E_n, h_n, \sigma_n, \mathcal{H}$	Canonical event record, chain head, optional signature, and hash-chain update.
$\mathfrak{N}_{[t_0,t_1]}$	Complete Version 0.3 bounded transition over a requested interval.

B

Proof and property sketches

B.1 Non-degeneracy

If one global differentiable F is identically zero, every directional derivative is zero. No normal, separation, or restoring direction can be obtained. Therefore any normal-based or conservative-advancement implementation needs nontrivial local fields, constraint Jacobians, support functions, feature distances, or multiple charts.

B.2 First-order boundary contraction

Let $g = \nabla F(z)$ and $z^+ = z - \lambda F(z)g/(\|g\|^2 + \varepsilon)$. Taylor expansion gives

$$F(z^+) = F(z)\left(1 - \lambda \frac{\|g\|^2}{\|g\|^2 + \varepsilon}\right) + O(\|z^+ - z\|^2). \tag{B.1}$$

For small residual and $0 < \lambda < 2$, the leading term contracts when the gradient is well conditioned. This property is local projection, not a proof of CCD over an interval.

B.3 Log-polar acceleration

With $r = r_0 b^w$, $\dot{r} = (\ln b)r\dot{w}$, and $\ddot{r} = r((\ln b)\ddot{w} + (\ln b)^2\dot{w}^2)$. Substituting into

$$a = (\ddot{r} - r\dot{\theta}^2)e_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})e_\theta \tag{B.2}$$

shows that scale, centripetal, and radial–angular cross terms cannot be represented by angle or independent bit shifts alone.

B.4 Möbius inversion in log-polar coordinates

For $z = re^{i\theta}$, $1/z = r^{-1}e^{-i\theta}$. Taking $w = \log_b(r/r_0)$ yields $w' = -w$ up to the declared reference-scale constant and $\theta' = -\theta$. Applying the map twice returns the original state away from the pole.

B.5 Swept-bound broad-phase completeness

Let $S_A(t)$ and $S_B(t)$ be the exact occupied sets expanded by the required margin. Suppose the broad phase uses bounds $\widehat{S}_A, \widehat{S}_B$ such that

$$\bigcup_{t\in[t_0,t_1]} S_A(t) \subseteq \widehat{S}_A, \qquad \bigcup_{t\in[t_0,t_1]} S_B(t) \subseteq \widehat{S}_B. \tag{B.3}$$

If a collision occurs, some point belongs to both exact swept sets and therefore to both bounds, so $\widehat{S}_A \cap \widehat{S}_B \neq \varnothing$. Thus pruning only disjoint conservative swept bounds cannot remove a true pair. The result fails immediately if either bound is not conservative or if the pair is dropped by capacity/filter logic.

B.6 Conservative advancement with a separation margin

Let $g(t)$ be a lower bound on pair separation minus $d_{\min}$, with $g(t) > 0$ and closing rate $-\dot{g}(t) \leq B$ over the candidate step. For

$$\Delta t \leq \eta \frac{g(t)}{B}, \qquad 0 < \eta < 1, \tag{B.4}$$

the worst possible decrease is $B\Delta t \leq \eta g(t) < g(t)$, so g cannot become nonpositive during that step. Repeating the argument preserves non-crossing until a contact bracket/tolerance is reached, provided the lower-bound and rate contracts remain valid.

B.7 Rigid angular closing-speed bound

For a point x on a rigid body with reference point c, rotational velocity is $\omega \times (x - c)$, hence

$$\|\omega \times (x - c)\| \leq \|\omega\| \, \|x - c\| \leq \|\omega\| R. \tag{B.5}$$

For two bodies, adding translational relative speed and the two angular terms gives a conservative scalar closing-speed bound. It may be loose, but omitting it is unsound for rotating bodies.

B.8 Analytic sphere–sphere TOI

For center difference $p(t) = p_0 + vt$ and combined radius/margin R, contact satisfies

$$\|p_0 + vt\|^2 - R^2 = at^2 + bt + c = 0, \tag{B.6}$$

where $a = v \cdot v$, $b = 2p_0 \cdot v$, and $c = p_0 \cdot p_0 - R^2$. Initial overlap is $c < 0$; no relative motion is $a = 0$; the earliest interval root is the first nonnegative root in the query interval. A stable quadratic implementation and interval post-check are still required in floating point.

B.9 Inclusion subdivision soundness

Suppose an interval extension $\Box f(I)$ contains $f(t)$ for every $t \in I$. If $0 \notin \Box f(I)$, no root of f lies in I, so the interval may be discarded. If $0 \in \Box f(I)$, subdivision/refinement preserves all possible roots. Therefore an inclusion solver that discards only zero-free boxes cannot miss a root, subject to a correct inclusion extension and complete queue processing.

B.10 Earliest-event interval reduction

For result intervals $I_k = [l_k, u_k]$, any candidate with $l_k > \min_j u_j + \tau_c$ cannot be the earliest event within the co-time tolerance. Retaining all intervals whose lower endpoint overlaps the smallest upper endpoint plus tolerance forms a conservative earliest bucket. Ordering only by midpoint does not have this property.

B.11 Bounded world-step progress

Each event loop either (a) advances logical time by at least a declared progress floor, (b) transitions a contact to a persistent/resting model that removes it from repeated impact localization, or (c) increments a finite event counter and terminates with failure at its bound. Under these rules the world step cannot execute an unbounded hidden loop.

B.12 Bounded memory

If every allocation is drawn from finite pools, every proposal is accepted only when capacity exists, and overflow returns failure or invokes a complete bounded fallback, memory remains below the declared sum by induction. Bounded memory alone does not imply complete CCD; the overflow result is part of correctness.

B.13 Parity detection and causal ring buffer

XOR parity changes under an odd number of bit inversions and remains unchanged under an even number; it is an incomplete integrity mechanism. If each physical timeline slot stores generation g, a read for generation g_n rejects any slot tagged above g_n, so wraparound reuses storage without violating logical order.

B.14 Hash-chain tamper evidence

If H is collision resistant and canonical records are unambiguous, changing an earlier event changes its digest input and every later chain head except with negligible collision probability. Trust still depends on protected keys/heads and truthful measurement; the chain does not prove the geometry or CCD bounds were correct.

C

Traceability to supplied sources and external CCD context

C.1 Supplied-source traceability

Source location	Source-derived development	Version 0.3 destination
[**S0**], pp. 1–6	Original vocabulary, kinematics, parity-conditioned boundary restoration.	Semantic commitments, local zero sets, parity/phase control, and closed NHDF chain.
[**S0**], pp. 8–18	Operator precedence, SDF moved into LUT, distributed apex, NHDF/IHCP naming.	Paradigm identity, typed operator algebra, local-zero-set substrate, and bounded runtime.
[**S0**], pp. 19–30	$\Delta^2\phi$, forward timeline, optical/quantum, SAR, and GPU framing.	Phase/timeline chapters, operator-equivalence extension, adapters, and GPU profile.
[**S1**], all	Version 0.1 non-degenerate formal model and evidence boundary.	Retained throughout Version 0.3 as the baseline representation/dynamics/runtime split.
[**S2**], pp. 1–4 (07:50–07:59)	Time edge, radial scale/Jacobian, and radial–angular coupling.	Conformal log-polar kinematics and apex regularization.
[**S2**], pp. 5–8 (08:03–08:13)	Möbius/gyrovector/spiral ideas, frame drift, phase lock, and temporal guard.	Declared transform profiles, observer frame, phase control, and workload guard, with corrected semantics.
[**S2**], pp. 8–10 (08:16)	Focal attenuation, interrupt, and coherence recovery.	Budgeted attention and interrupt chapter.
[**S2**], pp. 10–17 (08:19–08:25)	Literal graphics CCD intuition, SDF step check, sign crossing, sliding, damping, and fragmentation.	Starting requirement for the Version 0.3 CCD subsystem and response chapter; single-step/zero-overhead claims are not retained as proof.
[**S2**], pp. 17–21 (08:29–08:34)	Foveal concentration, dark/bright transition, and narrow visual mask.	Input conditioning, photometric shock, and attention profiles.
[**S2**], pp. 21–23 (08:37)	Siren, Doppler, acoustic interrupt.	Acoustic wavefront/Doppler adapter and public-safety profile.
[**S2**], pp. 23–25 (08:40–08:42)	Soft fields, stationary origin, and acceleration/deceleration discussion.	Field/motion contracts and explicit correction that log addressing does not replace physical acceleration.
[**S2**], pp. 26–30 (08:44–08:50)	Blackboard metadata, identity salt, and one-dimensional timeline.	Tamper-evident provenance with corrections separating identity from cryptographic key material.
[**S2**], pp. 31–40 (08:52–09:03)	Edge applications, police use cases, "digital jiu-jitsu," anisotropic mask, and state commit.	Application/public-safety profiles with human authority, privacy, no-autonomous-force, and evidence limits.

C.2 What the supplied sources do not establish

The supplied transcripts motivate an SDF distance check, endpoint sign transition, parity-triggered response, and the belief that CCD might become a register-level operation. They do not supply a general-purpose broad phase, shape/motion capability system, rigid rotational bounds, complete triangle feature tests, robust root isolation, deformable/self-collision handling, simultaneous-contact semantics, or numerical certification. Those additions are Version 0.3 engineering work informed by the external technical literature below, not facts attributed to the commute transcript.

C.3 External CCD research mapping

Technical context	**Use in Version 0.3**
Rigid conservative advancement and controlled advancement [1, 2]	Separation-based advancement, rigid angular bounds, capability limits, and conservative hit/clear semantics.
Hierarchy-based rigid/polygonal collision [6]	Conservative BVH broad phase and temporal hierarchy requirements.
Articulated, deformable, and volume CCD [3, 4, 5]	Motion classes, feature/volume backends, self-collision, and deformation envelopes.
Robust exact/inclusion CCD [9, 10, 11, 12, 13, 14, 15]	Interval/inclusion root isolation, exact/filtered predicates, degeneracy policy, and TOI enclosures.
Cloth and simultaneous collision handling [7, 8]	Persistent contact, co-time buckets, manifold/island construction, and event-order concerns.
Scalable parallel CCD [16]	Backend queues, deterministic/parallel reductions, datasets, and scaling telemetry.
Intersection-free/barrier dynamics [17, 18, 19]	Optional barrier-coupled response profile and strict separation between detection and response.

D

Source notes and version history

Source notes

[S0] 30-page exported prompt-and-response transcript beginning with "1bit parity bit lower case phi jitter log encoded polar LUT...", supplied by the author.

[S1] *Non-Euclidean Holographic Data Fields: A Formal Operator Specification for a Parity-Conditioned Kinematic Foliation and Implicit Holographic Co-Processor*, Version 0.1, Tom Klootwijk, 31 August 2026.

[S2] 40-page exported commute transcript beginning at approximately 07:50 with boarding the tram at IJburg and ending at approximately 09:03 at the Hoofddorp work location, supplied by the author.

External context
The primary CCD publications listed in the research-reference chapter were consulted to extend the source-derived contact intuition into a general-purpose engineering profile. They do not imply endorsement of NHDF by their authors.

Version history

Version	Date	Summary
0.1	31 Aug 2026	First non-degenerate formalization of NHDF, PCKF, and IHCP; local zero sets, typed chain, bounded routing, GPU/optical/SAR extensions, and validation program.
0.2	31 Aug 2026	Roads Technology commute extension: conformal log-polar metric, transform profiles, observer/photometric conditioning, attention/interrupt, adaptive timeline, initial conservative CCD/response, acoustics, provenance, responsible public-safety profile, and expanded corrections.
0.3	31 Aug 2026	General-purpose literal graphics CCD attempt: typed query/results, conservative broad phase, capability dispatch, analytic/convex/mesh/deformable/implicit backends, certified TOI enclosures, degeneracy and initial-overlap policy, manifold/simultaneous/self-contact handling, event-driven response, bounded GPU queues, conformance ladder, adversarial verification, and executable elementary reference scaffold.

End of NHDF Formal Specification, Version 0.3